

Softmax is 1/2-Lipschitz: A tight bound across all ℓ_p norms

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Abstract

The softmax function is a basic operator in machine learning and optimization, used in classification, attention mechanisms, reinforcement learning, game theory, and problems involving log-sum-exp terms. Existing **robustness** guarantees of learning models and convergence analysis of optimization algorithms typically consider the softmax operator to have a Lipschitz constant of 1 with respect to the ℓ_2 norm. In this work, we prove that the softmax function is **contractive** with the Lipschitz constant 1/2, **uniformly** across all ℓ_p norms with $p \geq 1$. We also show that the local Lipschitz constant of softmax attains 1/2 for $p = 1$ and $p = \infty$, and for $p \in (1, \infty)$, the constant remains strictly below 1/2 and the supremum 1/2 is achieved only in the limit. To our knowledge, this is the first **comprehensive** norm-uniform analysis of softmax Lipschitz continuity. We demonstrate how the sharper constant directly improves a range of existing theoretical results on robustness and convergence. We further validate the sharpness of the 1/2 Lipschitz constant of the softmax operator through empirical studies on attention-based architectures (ViT, GPT-2, Qwen3-8B) and on stochastic policies in reinforcement learning.

1 Introduction

The softmax function has applications across diverse areas of machine learning and optimization. In classification models, model output logits are normalized using the softmax function to form a probability distribution (Goodfellow et al., 2016). In Transformer architectures, softmax **normalizes** attention scores to compute weighted combinations of input token features, enabling high-quality feature **refinement** (Vaswani et al., 2017). These transformer architectures have shown **state-of-the-art** performance in various applications in natural language processing (Tunstall et al., 2022), computer vision (Khan et al., 2022), reinforcement learning (Li et al., 2023), etc. Since softmax is the gradient of the log-sum-exp function, its properties are crucial for analyzing optimization algorithms (Gao & Pavel, 2017; Nachum et al., 2017; Nesterov, 2005). In reinforcement learning, softmax is used to **convert** action-value estimates (Q-values) to stochastic policies. This results in a probability distribution over actions that favours actions with higher Q-values and still assigns probabilities to **suboptimal** actions. In **entropy-regularized** reinforcement learning, the softmax operator arises naturally as the solution of the policy optimization problem (Sutton & Barto, 2018; Nachum et al., 2017).

The softmax operator maps any real vector to a probability distribution. Formally, the softmax function $\sigma_\lambda: \mathbb{R}^n \rightarrow \Delta_n^\circ$ with **inverse-temperature** parameter $\lambda > 0$ is defined as

$$\sigma_\lambda(\mathbf{x})_i = \frac{\exp(\lambda x_i)}{\sum_{j=1}^n \exp(\lambda x_j)}, \quad i = 1, \dots, n, \quad (1)$$

where $\Delta_n = \{\mathbf{u} \in \mathbb{R}^n: u_i \geq 0, \sum_i u_i = 1\}$ is the probability simplex and $\Delta_n^\circ = \{\mathbf{u} \in \mathbb{R}^n: u_i > 0, \sum_i u_i = 1\}$ denotes its interior. Since $\exp(x_i) > 0$ for all $x_i \in \mathbb{R}$, $\sigma_\lambda(\mathbf{x})$ never attains the boundary of the simplex $\partial\Delta_n = \{\mathbf{u} \in \mathbb{R}^n: u_i = 0 \text{ for some } i\}$. The **coefficient** λ measures the smoothness of the output distribution. A larger λ makes the distribution more peaked on the largest entries, and a smaller λ makes it smoother with more **evenly** spread probability values. The standard softmax function is when $\lambda = 1$.

Softmax是 1/2-Lipschitz：在所有 ℓ_p 范数下均成立的一个紧致界限

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摘要

softmax函数是机器学习和优化中的基础算子，用于分类、注意力机制、强化学习、博弈论以及涉及log-sum-exp项的问题。现有的学习模型的鲁棒性保证和优化算法的收敛性分析通常考虑softmax算子相对于 ℓ_2 范数的Lipschitz常数为1。在本工作中，我们证明了softmax函数是收缩映射，其Lipschitz常数为 1/2，在所有满足 $p \geq 1$ 的 ℓ_p 范数下均成立。我们还表明，softmax的局部Lipschitz常数为 1/2，当 $p = 1$ 和 $p = \infty$ 时，对于 $p \in (1, \infty)$ ，该常数严格低于 1/2，其上确界 1/2 仅在极限中达到。据我们所知，这是首次对softmax Lipschitz连续性进行全面的范数统一分析。我们展示了更精确的常数如何直接改进现有关于鲁棒性和收敛性的理论结果。我们进一步通过在基于注意力的架构（ViT、GPT-2、Qwen3-8B）和强化学习中的随机策略上的实证研究，验证了softmax算子的Lipschitz常数的尖锐性。

1 引言

softmax函数在机器学习和优化等多个领域都有应用。在分类模型中，模型输出logits通过softmax函数进行归一化，形成概率分布（Goodfellow等人，2016年）。在Transformer架构中，softmax函数对注意力分数进行归一化，以计算输入token特征的加权组合，从而实现高质量的特征精炼（Vaswani等人，2017年）。这些Transformer架构在各种自然语言处理（Tunstall等人，2022年）、计算机视觉（Khan等人，2022年）、强化学习（Li等人，2023年）等应用中展现了最先进的性能。由于softmax是log-sum-exp函数的梯度，其性质对于分析优化算法至关重要（高 & 佩夫尔，2017年；Nachum等人，2017年；Nesterov，2005年）。在强化学习中，softmax用于将动作值估计（Q值）转换为随机策略。这会生成一个动作的概率分布，优先考虑Q值较高的动作，同时仍然为次优动作分配概率。在熵正则化强化学习中，softmax算子自然地作为策略优化问题的解出现（Sutton & Barto，2018年；Nachum等人，2017年）。

softmax算子将任何实向量映射为概率分布。形式上，softmax函数 $\sigma_\lambda: \mathbb{R}^n \rightarrow \Delta_n^\circ$ 具有逆温度参数 $\lambda > 0$ ，其定义为

$$\sigma_\lambda(\mathbf{x})_i = \frac{\exp(\lambda x_i)}{\sum_{j=1}^n \exp(\lambda x_j)}, \quad i = 1, \dots, n, \quad (1)$$

其中 $\Delta_n = \{\mathbf{u} \in \mathbb{R}^n: u_i \geq 0, \sum_i u_i = 1\}$ 是概率单纯形，而 $\Delta_n^\circ = \{\mathbf{u} \in \mathbb{R}^n: u_i > 0, \sum_i u_i = 1\}$ 表示其内部。由于 $\exp(x_i) > 0$ 对所有 $x_i \in \mathbb{R}$ ， $\sigma_\lambda(\mathbf{x})$ 从不达到单纯形的边界 $\partial\Delta_n = \{\mathbf{u} \in \mathbb{R}^n: u_i = 0 \text{ 对于某些 } i\}$ 。系数 λ 衡量输出分布的光滑度。更大的 λ 使分布在最大值上更尖锐，而更小的 λ 使它更平滑，概率值更均匀地分布。标准的softmax函数是当 $\lambda = 1$ 时。

Accurate analysis of softmax’s Lipschitz constant has important applications across machine learning and optimization. It enables accurate robustness analysis of attention mechanisms, where Lipschitz bounds can be used to set hyperparameters that ensure stable training (Qi et al., 2023). In game-theoretic learning, modeling action selection via softmax allows Lipschitz continuity to be directly **leveraged for** establishing convergence rates toward Nash equilibria (Gao & Pavel, 2017). Many existing generalization results for learning models require computing the Lipschitz constant of the loss function with respect to either network input or network parameters. When softmax appears in the final layer or in attention, an accurate Lipschitz analysis becomes essential for deriving sharp **generalization bounds** (Asadi & Abbe, 2020). Moreover, when a model is expressed as a composition of layers, standard network analyses bound the global Lipschitz constant by an appropriate composition (often a product) of layer-wise constants Bartlett et al. (2017); Miyato et al. (2018); Virmaux & Scaman (2018); Fazlyab et al. (2019). In such bounds, the softmax Lipschitz constant acts as a **multiplicative factor**, so tightening it leads directly to sharper network-level guarantees. In classification networks, robustness and domain adaptation depend on how sensitive output probabilities are to input **perturbations**, which is again controlled by the Lipschitz constant of softmax (Chen et al., 2022). The log-sum-exp (LSE) function is a smooth approximation of the **maximum operator**, and the gradient of LSE is the softmax function. Hence, in optimization problems with LSE-based objective functions, step-size rules for gradient descent and smoothness-based convergence rates can be sharpened using a tight softmax Lipschitz bound. Lipschitz constant of softmax is also used in analysing several other frameworks, including meta-learning (Jeon et al., 2024), structural **causal models** (Le Priol et al., 2021), prior-data fitted networks (Nagler, 2023), and **sparse training** (Lei et al., 2024). The above-mentioned diverse set of applications highlights that deriving a tight Lipschitz constant of softmax has a substantial practical impact.

Despite the theoretical and empirical significance of softmax’s Lipschitz constant, an accurate analysis is missing in the literature. A number of works (Gao & Pavel, 2017; Laha et al., 2018; Asadi & Abbe, 2020; Gouk et al., 2021; Le Priol et al., 2021; Chen et al., 2022; Nagler, 2023; Jeon et al., 2024; Lei et al., 2024) assume the softmax function to be 1-Lipschitz with respect to the ℓ_2 norm. This assumption is often attributed to Proposition 4 in Gao & Pavel (2017). In some works, this is mentioned as a straightforward fact, perhaps because showing the Lipschitz constant to be bounded by 1 is trivial. We also note that Xu et al. (2022) reports a bound of 1/4 for the softmax’s Lipschitz constant. The main motivation for this work is the discrepancy between the commonly assumed Lipschitz constant of the softmax function and the smaller values observed in our empirical studies on transformer architectures. This prompted us to derive a tighter bound for the Lipschitz constant of the softmax function. In this regard, our contributions are as follows:

- We prove that softmax is 1/2-Lipschitz across all ℓ_p norms ($p \geq 1$), thereby improving the usually assumed bound of 1.
- Furthermore, we show the tightness of this bound by proving that the local Lipschitz constant is attained for $p = 1$ and $p = \infty$ as 1/2, while for $p \in (1, \infty)$ the local Lipschitz constants remain strictly below 1/2 and the supremum is only approached in the limit.
- We **demonstrate** that our tight Lipschitz bound improves the existing theoretical results in robustness and convergence. We also validate our derived Lipschitz constant through experiments on large-scale transformer models and reinforcement learning policies.

2 Preliminaries

In this section, we provide the necessary background for our main results. We start with the definition of the ℓ_p norm for vectors and the **corresponding** induced operator norm for matrices.

Definition 2.1 (ℓ_p norm). For a vector $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ and for $1 \leq p < \infty$, the ℓ_p norm of $\mathbf{x}$ is defined as

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

对softmax的利普希茨常数进行精确分析在机器学习和优化领域具有重要应用。它能够实现对注意力机制精确的鲁棒性分析，其中利普希茨界可用于设置确保稳定训练的超参数（Qi等人，2023年）。在博弈学习理论中，通过softmax建模动作选择允许直接利用利普希茨连续性来建立向纳什均衡的收敛速度（高 & 佩夫尔，2017年）。许多现有的学习模型泛化结果需要计算损失函数相对于网络输入或网络参数的利普希茨常数。当softmax出现在最终层或注意力机制中时，进行精确的利普希茨分析对于推导出尖锐的泛化界至关重要（Asadi & Abbe，2020年）。此外，当模型表示为层级的组合时，标准的网络分析通过适当的层级常数组合（通常是乘积）来界定全局利普希茨常数（Bartlett等人，2017年；Miyato等人，2018年；Virmaux & Scaman，2018年；Fazlyab等人，2019年）。在这种界限中，softmax的利普希茨常数作为乘法因子，因此收紧它直接导致更尖锐的网络级保证。在分类网络中，鲁棒性和领域自适应取决于输出概率对输入扰动的敏感性，这再次由softmax的利普希茨常数控制（Chen等人，2022年）。对数和指数（LSE）函数是最大算子的平滑近似，LSE的梯度是softmax函数。因此，在具有基于LSE的目标函数的优化问题中，可以使用紧密的softmax利普希茨界来锐化梯度下降的步长规则和基于平滑性的收敛速度。softmax的利普希茨常数也用于分析其他框架，包括元学习（Jeon等人，2024年）、结构因果模型（Le Priol等人，2021年）、先验数据拟合网络（Nagler，2023年）和稀疏训练（Lei等人，2024年）。上述各种应用突出了推导出紧密的softmax利普希茨常数具有实质性的实际影响。

尽管softmax的利普希茨常数具有理论和实践上的重要意义，但文献中缺乏精确的分析。许多工作（高 & 佩夫尔，2017年；拉哈等，2018年；阿萨迪 & 阿贝，2020年；古克等，2021年；勒普里奥等，2021年；陈等，2022年；纳格尔，2023年；全等，2024年；雷等，2024年）假设softmax函数关于 ℓ_2 范数是1-利普希茨的。这一假设通常归因于高 & 佩夫尔（2017年）中的命题4。在一些工作中，这被提及为一个显而易见的事实，也许因为证明利普希茨常数被1界定是平凡的。我们还注意到，徐等（2022年）报告了softmax的利普希茨常数的一个 1/4 界限。本工作的主要动机是softmax函数的通常假设的利普希茨常数与我们关于Transformer架构的实证研究中观察到的较小值之间的差异。这促使我们为softmax的利普希茨常数推导一个更紧密的界限。在这方面，我们的贡献如下：

- 我们证明了softmax在所有 ℓ_p 范数($p \geq 1$)下是 1/2-Lipschitz的，从而改进了通常假设的1界。
- 此外，我们通过证明局部Lipschitz常数在 $p = 1$ 和 $p = \infty$ 处达到 1/2，而在 $p \in (1, \infty)$ 处局部 Lipschitz常数严格低于 1/2，并且只有在极限中才达到 supremum，来证明此界的紧致性。
- 我们证明了我们的紧致Lipschitz界改进了现有在鲁棒性和收敛性方面的理论结果。我们还通过在大规模Transformer模型和强化学习策略上的实验验证了我们推导出的Lipschitz常数。

2 预备知识

在本节中，我们提供了我们主要结果所需的背景。我们从向量的 ℓ_p 范数的定义以及相应的矩阵的诱导算子范数开始。

定义 2.1 (ℓ_p 范数)。对于向量 $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ 以及 $1 \leq p < \infty$ ， ℓ_p 范数 $\mathbf{x}$ 定义为

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

For $p = \infty$, the ℓ_∞ norm is defined as

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

We can also define the operator norm of a matrix $\mathbf{A} = (A_{ij}) \in \mathbb{R}^{m \times n}$ induced by the ℓ_p norm as

$$\|\mathbf{A}\|_p := \sup_{\mathbf{v} \neq 0} \frac{\|\mathbf{A}\mathbf{v}\|_p}{\|\mathbf{v}\|_p} = \sup_{\|\mathbf{v}\|_p=1} \|\mathbf{A}\mathbf{v}\|_p.$$

Throughout the paper, we overload $\|\cdot\|_p$ to represent both the vector norm and the corresponding induced norm on matrices. Next, we define the Lipschitz property of functions.

Definition 2.2 (Lipschitz continuity in ℓ_p norm). Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and let $\|\cdot\|_p$ denote the ℓ_p norm with $1 \leq p \leq \infty$. The function f is said to be Lipschitz continuous with respect to $\|\cdot\|_p$ if there exists a constant $L_p \geq 0$ such that

$$\|f(\mathbf{x}) - f(\mathbf{y})\|_p \leq L_p \|\mathbf{x} - \mathbf{y}\|_p, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n. \quad (2)$$

The smallest such constant L_p is called the Lipschitz constant of f with respect to the ℓ_p norm.

A mapping f is **contractive** if $L_p < 1$, and **non-expansive** if $L_p \leq 1$. It is firmly non-expansive if $\|f(\mathbf{x}) - f(\mathbf{y})\|_p^2 \leq \langle f(\mathbf{x}) - f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y}$, and co-coercive with constant $\beta > 0$ if $\langle f(\mathbf{x}) - f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq \beta \|f(\mathbf{x}) - f(\mathbf{y})\|_p^2$.

Note that the Lipschitz constant in Eq. 2 is a global quantity, since the inequality holds for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. By definition, L_p can be characterized as,

$$L_p = \sup_{\mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{y}} \frac{\|f(\mathbf{x}) - f(\mathbf{y})\|_p}{\|\mathbf{x} - \mathbf{y}\|_p}. \quad (3)$$

We next introduce the notion of a local Lipschitz constant (Rockafellar & Wets, 1998) via the Jacobian of f , which in turn provides another **characterization** of the global Lipschitz constant.

Definition 2.3 (Jacobian matrix (Boyd & Vandenberghe, 2004)). Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a differentiable function, where $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$ and each $f_i: \mathbb{R}^n \rightarrow \mathbb{R}$ is scalar-valued function. The Jacobian matrix of f at $\mathbf{x}$, denoted $\mathbf{J}_f(\mathbf{x}) \in \mathbb{R}^{m \times n}$, is the matrix of all partial derivatives,

$$\mathbf{J}_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}) \end{bmatrix}.$$

If the function is **continuously differentiable**, the local Lipschitz constant of a function at any $\mathbf{x} \in \mathbb{R}^n$ is equal to the ℓ_p norm of the Jacobian at that point.

Definition 2.4 (Local Lipschitz constant (Rockafellar & Wets, 1998)). Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$. The local Lipschitz constant of f at a point $\mathbf{x} \in \mathbb{R}^n$ with respect to the ℓ_p norm is defined as

$$L_p(\mathbf{x}) := \limsup_{\mathbf{y} \rightarrow \mathbf{x}, \mathbf{y} \neq \mathbf{x}} \frac{\|f(\mathbf{y}) - f(\mathbf{x})\|_p}{\|\mathbf{y} - \mathbf{x}\|_p}.$$

Intuitively, $L_p(\mathbf{x})$ characterizes the tightest Lipschitz bound in an **arbitrarily** small neighborhood of $\mathbf{x}$. If f is differentiable at $\mathbf{x}$, then

$$L_p(\mathbf{x}) = \|\mathbf{J}_f(\mathbf{x})\|_p,$$

where $\|\mathbf{J}_f(\mathbf{x})\|_p$ denotes ℓ_p -induced operator norm of the Jacobian matrix.

We now relate the global Lipschitz constant of a function to its local Lipschitz constant, following a result from Hytönen et al. (2016).

Lemma 1 (Lipschitz constant via the Jacobian). *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be continuously differentiable. Then, for $1 \leq p \leq \infty$, the global Lipschitz constant of f with respect to ℓ_p norm is given as*

$$L_p = \sup_{\mathbf{x} \in \mathbb{R}^n} L_p(\mathbf{x}) = \sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_f(\mathbf{x})\|_p.$$

对于 $p = \infty$, ℓ_∞ 范数定义为

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

我们也可以定义矩阵 $\mathbf{A} = (A_{ij}) \in \mathbb{R}^{m \times n}$ 由 ℓ_p 范数诱导的算子范数

$$\|\mathbf{A}\|_p := \sup_{\mathbf{v} \neq 0} \frac{\|\mathbf{A}\mathbf{v}\|_p}{\|\mathbf{v}\|_p} = \sup_{\|\mathbf{v}\|_p=1} \|\mathbf{A}\mathbf{v}\|_p.$$

全文中, 我们重载 $\|\cdot\|_p$ 来表示向量范数和矩阵上相应的诱导范数。接下来, 我们定义函数的 Lipschitz 性质。

定义 2.2 (ℓ_p 范数中的 Lipschitz 连续性)。设 $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, 且 $\|\cdot\|_p$ 表示具有 $1 \leq p \leq \infty$ 的 ℓ_p 范数。如果函数 f 关于 $\|\cdot\|_p$ 存在一个常数 $L_p \geq 0$ 使得

$$\|f(\mathbf{x}) - f(\mathbf{y})\|_p \leq L_p \|\mathbf{x} - \mathbf{y}\|_p, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n. \quad (2)$$

使该不等式成立的最小常数 L_p 称为 f 关于 ℓ_p 范数的 Lipschitz 常数。

一个映射 f 是收缩映射当且仅当 $L_p < 1$, 是非扩张映射当且仅当 $L_p \leq 1$ 。它是强非扩张映射当且仅当 $\|f(\mathbf{x}) - f(\mathbf{y})\|_p^2 \leq \langle f(\mathbf{x}) - f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$ 对于所有 $\mathbf{x}, \mathbf{y}$, 并且具有常数 $\beta > 0$ 使得 $\langle f(\mathbf{x}) - f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq \beta \|f(\mathbf{x}) - f(\mathbf{y})\|_p^2$ 对于所有 $\mathbf{x}, \mathbf{y}$ 。共 coercive 如果 $\langle f(\mathbf{x}) - f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq 0$ 对于所有 $\mathbf{x}, \mathbf{y}$ 。

请注意, 方程式 2 中的利普希茨常数是一个全局量, 因为该不等式对所有 $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ 都成立。根据定义, L_p 可以被描述为,

$$L_p = \sup_{\mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{y}} \frac{\|f(\mathbf{x}) - f(\mathbf{y})\|_p}{\|\mathbf{x} - \mathbf{y}\|_p}. \quad (3)$$

接下来, 我们通过 f 的雅可比矩阵引入局部利普希茨常数的概念 (Rockafellar & Wets, 1998), 这反过来又为全局利普希茨常数提供了另一种表征。

定义 2.3 (雅可比矩阵 (Boyd & Vandenberghe, 2004))。设 $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ 是一个可微函数, 其中 $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$, 且每个 $f_i: \mathbb{R}^n \rightarrow \mathbb{R}$ 是标量值函数。函数 f 在 $\mathbf{x}$ 处的雅可比矩阵, 记作 $\mathbf{J}_f(\mathbf{x}) \in \mathbb{R}^{m \times n}$, 是由所有偏导数组成的矩阵,

$$\mathbf{J}_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}) \end{bmatrix}.$$

如果函数是 **该点处雅可比矩阵的 ℓ_p 范数**。到 ℓ_p 范数 **该点处雅可比矩阵的 $\{\ell_p\}$ 范数**。

定义 2.4 (局部 Lipschitz 常数 (Rockafellar & Wets, 1998))。设 $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ 。函数 f 在点 $\mathbf{x} \in \mathbb{R}^n$ 处关于 ℓ_p 范数的局部 Lipschitz 常数定义为

$$L_p(\mathbf{x}) := \limsup_{\mathbf{y} \rightarrow \mathbf{x}, \mathbf{y} \neq \mathbf{x}} \frac{\|f(\mathbf{y}) - f(\mathbf{x})\|_p}{\|\mathbf{y} - \mathbf{x}\|_p}.$$

直观上, $L_p(\mathbf{x})$ 刻画了在 $\mathbf{x}$ 附近任意小邻域内的最紧利普希茨界。如果 f 在 $\mathbf{x}$ 处可微, 那么

$$L_p(\mathbf{x}) = \|\mathbf{J}_f(\mathbf{x})\|_p,$$

其中 $\|\mathbf{J}_f(\mathbf{x})\|_p$ 表示雅可比矩阵的 ℓ_p -诱导算子范数。

我们现在将函数的全局 Lipschitz 常数与其局部 Lipschitz 常数联系起来, 遵循 Hytönen 等人 (2016) 的一个结果。

引理 1 (通过雅可比的利普希茨常数). *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be 连续可微. Then, for $1 \leq p \leq \infty$, the global Lipschitz constant of f with respect to ℓ_p norm is given as*

$$L_p = \sup_{\mathbf{x} \in \mathbb{R}^n} L_p(\mathbf{x}) = \sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_f(\mathbf{x})\|_p.$$

Thus, the global Lipschitz constant of a function can be expressed in a variational form involving the ℓ_p -induced operator norm of its Jacobian matrix. To apply this principle to the softmax operator, we recall its Jacobian formulation from Gao & Pavel (2017), as stated below.

Lemma 2 (Jacobian of the softmax). *Let $\sigma_\lambda : \mathbb{R}^n \rightarrow \Delta_n^\circ$ be the softmax function as in Definition 1 and let $\mathbf{s} = \sigma_\lambda(\mathbf{x})$. Then, the Jacobian of σ_λ at $\mathbf{x}$ is*

$$\mathbf{J}_{\sigma_\lambda}(\mathbf{x}) = \lambda (\text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top),$$

so, in coordinates, if $\mathbf{s} = \{s_1, s_2, \dots, s_n\}$,

$$[\mathbf{J}_{\sigma_\lambda}(\mathbf{x})]_{ij} = \begin{cases} \lambda s_i(1 - s_i), & i = j, \\ -\lambda s_i s_j, & i \neq j. \end{cases}$$

Refer to Proposition 2 in Gao & Pavel (2017) for the proof of Lemma 2. In Section 3, we will make use of Lemma 1 and the result in Lemma 2 for the softmax’s Jacobian to establish our main results.

3 Main Results

We begin by deriving an **inequality result** for ℓ_p -induced operator norms for matrices.

Proposition 1 (Norm Interpolation). *Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. Then:*

- (a) $\|\mathbf{A}\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |A_{ij}|$.
- (b) $\|\mathbf{A}\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |A_{ij}|$.
- (c) For $1 < p < \infty$,

$$\|\mathbf{A}\|_p \leq \|\mathbf{A}\|_1^{1/p} \|\mathbf{A}\|_\infty^{1-1/p}.$$

Proposition 1(a) and (b) state the well-known results that the ℓ_1 -induced operator norm of a matrix is equal to its maximum absolute column sum, and the ℓ_∞ -induced operator norm is equal to the maximum absolute row sum (Golub & Van Loan, 2013). Proposition 1(c) establishes an interpolation inequality that upper bounds the ℓ_p -induced operator norm of a matrix in terms of $\|\mathbf{A}\|_1$ and $\|\mathbf{A}\|_\infty$. While this inequality is a special case of the Riesz–Thorin interpolation theorem (Riesz, 1927; Stein, 1956; Thorin, 1939), we provide a self-contained proof in the Appendix A.1 for clarity. Importantly, Proposition 1(c) forms the key technical tool for this section, as it enables us to derive norm-uniform bounds on the Lipschitz constant of the softmax operator. In particular, we reformulate the optimization problem in Lemma 1 using the Jacobian form of the softmax provided in Lemma 2.

Lemma 3 (Lipschitz constant of the softmax operator). *Let $\sigma_\lambda : \mathbb{R}^n \rightarrow \Delta_n^\circ$ be the softmax function as in Definition 1, and let $\mathbf{s} = \sigma_\lambda(\mathbf{x})$. Then, for any $1 \leq p \leq \infty$, the Lipschitz constant of σ_λ with respect to ℓ_p norm is given by*

$$L_p = \lambda \sup_{\mathbf{s} \in \Delta_n^\circ} \|\text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top\|_p. \quad (4)$$

The key **implication** of Lemma 3 is that the global Lipschitz constant reduces to the supremum of the induced matrix norm in Eq. 4 over the interior of the probability simplex. The boundary points of the simplex, $\partial\Delta_n$, do not influence the **formulation** of the Lipschitz constant, since all components of the softmax output are strictly positive.

Next, we combine the interpolation inequality from Proposition 1 with the Jacobian formulation of the softmax operator from Lemma 3 to establish its Lipschitz constant across all ℓ_p norms. This result is summarized as the main contribution of our work in Theorem 1.

Theorem 1 (Lipschitz constant of softmax function). *Irrespective of the ℓ_p norms ($p \geq 1$),*

- (a) $\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p \leq \frac{1}{2}$
- (b) $\|\sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y})\|_p \leq \frac{\lambda}{2} \|\mathbf{x} - \mathbf{y}\|_p$

因此，一个函数的全局Lipschitz常数可以表示为涉及其雅可比矩阵的 ℓ_p -诱导算子范数的变分形式。要将此原理应用于softmax算子，我们回忆Gao & 佩夫尔（2017）中给出的其雅可比公式，如下所述。

引理2 (softmax的雅可比). 令 $\sigma_\lambda : \mathbb{R}^n \rightarrow \Delta_n^\circ$ 为定义1中的 **softmax** 函数，并令 $\mathbf{s} = \sigma_\lambda(\mathbf{x})$ 。那么， σ_λ 在 $\mathbf{x}$ 处的雅可比为

$$\mathbf{J}_{\sigma_\lambda}(\mathbf{x}) = \lambda (\text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top),$$

因此，在坐标表示下，如果 $\mathbf{s} = \{s_1, s_2, \dots, s_n\}$,

$$[\mathbf{J}_{\sigma_\lambda}(\mathbf{x})]_{ij} = \begin{cases} \lambda s_i(1 - s_i), & i = j, \\ -\lambda s_i s_j, & i \neq j. \end{cases}$$

引理2的证明可参考高 & 佩夫尔（2017）中的命题2。在第3节，我们将利用引理1和引理2中关于 softmax 雅可比的结果，来建立我们的主要结果。

3 主要结果

我们首先推导出矩阵 ℓ_p -诱导算子范数的不等式结果。

命题1 (范数插值)。令 $\mathbf{A} \in \mathbb{R}^{n \times n}$ 。则：

- (a) $\|\mathbf{A}\|_1 = \text{最大}_{1 \leq j \leq n} \sum_{i=1}^n |A_{ij}|$.
- (b) $\|\mathbf{A}\|_\infty = \text{最大}_{1 \leq i \leq n} \sum_{j=1}^n |A_{ij}|$.
- (c) 对于 $1 < p < \infty$,

$$\|\mathbf{A}\|_p \leq \|\mathbf{A}\|_1^{1/p} \|\mathbf{A}\|_\infty^{1-1/p}.$$

命题1(a)和(b)陈述了众所周知的结论：矩阵的 ℓ_1 -诱导算子范数等于其最大绝对列和，而 ℓ_∞ -诱导算子范数等于最大绝对行和（Golub & Van Loan, 2013）。命题1(c)建立了一个插值不等式，该不等式以 $\|\mathbf{A}\|_1$ 和 $\|\mathbf{A}\|_\infty$ 为参数，上界了矩阵的 ℓ_p -诱导算子范数。虽然这个不等式是里斯-索林插值定理（里斯，1927；斯坦，1956；索林，1939）的一个特例，但我们在附录A.1中提供了一个自包含的证明，以保持清晰。重要的是，命题1(c)构成了本节的关键技术工具，因为它使我们能够推导出softmax算子的利普希茨常数的一致界。特别是，我们使用引理2中给出的softmax的雅可比形式，重新表述了引理1中的优化问题。

引理3 (softmax算子的Lipschitz常数). 令 $\sigma_\lambda : \mathbb{R}^n \rightarrow \Delta_n^\circ$ 为定义1中的**softmax**函数，并令 $\mathbf{s} = \sigma_\lambda(\mathbf{x})$ 。那么，对于任何 $1 \leq p \leq \infty$ ， σ_λ 关于 ℓ_p 范数的**Lipschitz**常数由下式给出

$$L_p = \lambda \sup_{\mathbf{s} \in \Delta_n^\circ} \|\text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top\|_p. \quad (4)$$

引理3的关键推论是，全局Lipschitz常数简化为公式4中概率单纯形内部诱导矩阵范数的上确界。单纯形的边界点 $\partial\Delta_n$ 不会影响Lipschitz常数的公式，因为softmax输出的所有分量都是严格正的。

接下来，我们将命题1中的插值不等式与引理3中softmax算子的雅可比公式相结合，以建立其在所有 ℓ_p 范数下的Lipschitz常数。这一结果总结为我们的工作的主要贡献，即定理1。

定理1 (softmax函数的利普希茨常数). 无论 ℓ_p 范数 ($p \geq 1$),

- (a) $\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p \leq \frac{1}{2}$
- (b) $\|\sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y})\|_p \leq \frac{\lambda}{2} \|\mathbf{x} - \mathbf{y}\|_p$

The upper bound on the Jacobian of the softmax operator across ℓ_p norms, established in Theorem 1, directly provides the global Lipschitz constant of the softmax function. Refer to Appendix A.2 for a detailed proof. A natural question arises that while many works assume the constant to be 1, we show it is in fact 1/2; but could it be even smaller, for instance 1/4 as claimed by Xu et al. (2022). Proposition 2 resolves this question by proving that $\lambda/2$ is indeed the tight global Lipschitz constant of the softmax operator for any $\lambda > 0$.

Proposition 2 (Tightness of the Lipschitz constant for softmax). *Consider the optimization problem,*

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p = \sup_{\mathbf{s} \in \Delta_n^\circ} \|\text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top\|_p,$$

where Δ_n° is the interior of the probability simplex.

- (a) For $p = 1$ or $p = \infty$, the supremum is $1/2$ and is attained at an interior point in Δ_n° . In particular, for $\mathbf{x} = (\ln(n-1), 0, 0, \dots, 0) \in \mathbb{R}^n$, the softmax output $\mathbf{s} = \sigma_1(\mathbf{x})$ satisfies $\mathbf{s} \in \Delta_n^\circ$ and

$$\|\text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top\|_p = \frac{1}{2}.$$

- (b) For $p \in (1, \infty)$, the supremum value is again $1/2$ but is not attained in Δ_n° for $n > 2$. Instead, there exists a sequence $\{\mathbf{s}_k\}_{k \geq 1} \subset \Delta_n^\circ$ converging to the boundary of the probability simplex $\partial\Delta_n$ such that

$$\lim_{k \rightarrow \infty} \|\text{Diag}(\mathbf{s}_k) - \mathbf{s}_k\mathbf{s}_k^\top\|_p = \frac{1}{2},$$

with $\mathbf{s}_k = \sigma_1(\mathbf{x}_k)$ for some $\mathbf{x}_k \in \mathbb{R}^n$. For $n = 2$, the supremum $\frac{1}{2}$ is attained at a point in Δ_2° .

The proof of Proposition 2 is deferred to Appendix A.3. In particular, part (b) relies on the fact that for $1 < p < \infty$, the interpolation inequality in Proposition 1 is indeed strict for $\mathbf{J}_{\sigma_1}(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$. This is because the supremum in the optimization problem of Proposition 2 can be attained only on the boundary points of the probability simplex. Specifically, the extremal points are the permutations of $(1/2, 1/2, 0, \dots, 0)$, which lie in $\partial\Delta_n$.

Remark. There are multiple works showing that the Lipschitz constant of the softmax operator with respect to the ℓ_2 norm is upper bounded by $1/2$. Alghamdi et al. (2022) upper bounds $\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_2$ by first controlling the Frobenius norm of the softmax Jacobian (using $\|\mathbf{A}\|_2 \leq \|\mathbf{A}\|_F$) and then maximising this bound over the simplex to obtain the same constant $1/2$. Yudin et al. (2025) analyse the Jacobian matrix $\mathbf{M}(\mathbf{s}) = \text{diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top$ on the simplex and upper bound its spectral norm via an eigenvalue argument (reducing the worst case to a distribution supported on two classes), yielding $\|\mathbf{M}(\mathbf{s})\|_2 \leq 1/2$. Similarly, Newhouse (2025) gives an independent proof by applying Gershgorin’s circle theorem to $\text{diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top$, and also shows tightness in the ℓ_2 setting. In contrast, we prove that the bound $1/2$ is tight uniformly across all ℓ_p norms for $p \geq 1$, and give a detailed analysis of attainability on the simplex for different p .

We next provide an example of a pair of points for which the empirically computed Lipschitz ratio approaches $1/2$. This gives empirical evidence for the tightness of our theoretical bound.

Example 1 (Tightness of the Lipschitz bound). *Let $K = 20$ and $\varepsilon = 10^{-4}$. Consider the vectors*

$$\mathbf{x} = (0, 0, -K, -K, \dots, -K) \in \mathbb{R}^{10}, \quad \mathbf{y} = \mathbf{x} + \varepsilon \mathbf{v},$$

where $\mathbf{v}$ is eigenvector corresponding to the maximum eigenvalue of $J_{\sigma_1}(\mathbf{x})$. Then the ratio

$$\frac{\|\sigma_1(\mathbf{y}) - \sigma_1(\mathbf{x})\|_p}{\|\mathbf{y} - \mathbf{x}\|_p}$$

evaluates approximately to 0.49999999504472 for all $p \geq 1$. This demonstrates a concrete pair $(\mathbf{x}, \mathbf{y})$ where the Lipschitz constant of the softmax function is nearly attained. We can extend the example to any dimension by adding $-K$ as a value to the other added dimensions.

定理1中建立的softmax算子关于 ℓ_p 范数的雅可比上界，直接给出了softmax函数的全局利普希茨常数。参考附录A.2的详细证明。一个自然的问题是，虽然许多工作假设该常数为1，但我们证明它实际上是 $1/2$ ；但它是否可以更小，例如Xu等人（2022）声称的 $1/4$ ？命题2通过证明 $\lambda/2$ 确实是softmax算子对任何 $\lambda > 0$ 的紧全局利普希茨常数，解决了这个问题。

命题 2 (softmax的利普希茨常数紧致性). 考虑优化问题

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p = \sup_{\mathbf{s} \in \Delta_n^\circ} \|\text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top\|_p,$$

其中 Δ_n° 是概率单纯形的内部。

- (a) 对于 $p = 1$ 或 $p = \infty$ ，上确界是 $1/2$ ，并在 Δ_n° 的内部点处达到。特别是，对于 $\mathbf{x} = (\ln(n-1), 0, 0, \dots, 0) \in \mathbb{R}^n$ ，softmax输出 $\mathbf{s} = \sigma_1(\mathbf{x})$ 满足 $\mathbf{s} \in \Delta_n^\circ$ 和

$$\|\text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top\|_p = \frac{1}{2}.$$

- (b) 对于 $p \in (1, \infty)$ ，上确界值仍然是 $1/2$ ，但在 Δ_n° 处对于 $n > 2$ 不达到。相反，存在一个序列 $\{\mathbf{s}_k\}_{k \geq 1} \subset \Delta_n^\circ$ 收敛到概率单纯形 $\partial\Delta_n$ 的边界，使得

$$\lim_{k \rightarrow \infty} \|\text{Diag}(\mathbf{s}_k) - \mathbf{s}_k\mathbf{s}_k^\top\|_p = \frac{1}{2},$$

对于某些 $\mathbf{x}_k \in \mathbb{R}^n$ ，使用 $\mathbf{s}_k = \sigma_1(\mathbf{x}_k)$ 。对于 $n = 2$ ，上确界 $1/2$ 在 Δ_2° 中的一个点上达到。

命题2的证明被推迟到附录A.3。特别是，部分(b)依赖于这样一个事实：对于 $1 < p < \infty$ ，命题1中的插值不等式对于 $\mathbf{J}_{\sigma_1}(\mathbf{x})$ 对于所有 $\mathbf{x} \in \mathbb{R}^n$ 确实是严格的。这是因为命题2的优化问题中的上确界只能在对概率单纯形的边界点上达到。具体来说，极值点是 $(1/2, 1/2, 0, \dots, 0)$ 的排列，它们位于 $\partial\Delta_n$ 。

备注。有多项工作表明，softmax算子相对于 ℓ_2 范数的利普希茨常数是上界为 $1/2$ 。Alghamdi等人 (2022) 首先通过控制softmax雅可比的Frobenius范数(使用 $\|\mathbf{A}\|_2 \leq \|\mathbf{A}\|_F$) 来上界 $\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_2$ ，然后通过单纯形上最大化这个界来得到相同的常数 $1/2$ 。Yudin等人(2025)分析单纯形上的雅可比矩阵 $\mathbf{M}(\mathbf{s}) = \text{diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top$ ，并通过特征值论证来上界其谱范数(将最坏情况减少到支持在两个类上的分布)，得到 $\|\mathbf{M}(\mathbf{s})\|_2 \leq 1/2$ 。类似地，Newhouse (2025) 通过将Gershgorin圆定理应用于 $\text{diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top$ 给出一个独立的证明，并且还表明在 ℓ_2 设置中是紧的。相比之下，我们证明了对于 $p \geq 1$ ，界 $1/2$ 在所有 ℓ_p 范数上都是统一的，并且在不同的 p 上给出了在单纯形上可达性的详细分析。

我们接下来给出一个例子，其中经验计算的利普希茨比率趋近于 $1/2$ 。这为我们的理论界提供了经验证据，证明其紧致性。

Example 1 (利普希茨界的紧致性). 令 $K = 20$ 和 $\varepsilon = 10^{-4}$ 。考虑向量

$$\mathbf{x} = (0, 0, -K, -K, \dots, -K) \in \mathbb{R}^{10}, \quad \mathbf{y} = \mathbf{x} + \varepsilon \mathbf{v},$$

其中 $\mathbf{v}$ 是对应于 J_{σ_1} 的最大特征值的特征向量。那么比率

$$\frac{\|\sigma_1(\mathbf{y}) - \sigma_1(\mathbf{x})\|_p}{\|\mathbf{y} - \mathbf{x}\|_p}$$

评估结果约为 0.49999999504472 对所有 $p \geq 1$ 都成立。这表明存在一个具体的对 $(\mathbf{x}, \mathbf{y})$ ，其中softmax函数的利普希茨常数几乎被达到。我们可以通过将 $-K$ 作为值添加到其他增加的维度来扩展此示例到任意维度。

4 Implications of the Improved Softmax Lipschitz Constant

Our result in Theorem 1 establishes that the softmax function is $\lambda/2$ -Lipschitz with respect to all ℓ_p norms for $p \geq 1$. This tighter characterization enables us to revisit existing results where softmax Lipschitz continuity is relevant, leading to sharper constants or simplified identities. Next, we illustrate this in a few representative settings.

4.1 Refinement of (Gao & Pavel, 2017, Cor. 3).

Leveraging our sharper Lipschitz estimate, Corollary 3 in Gao & Pavel (2017) can be strengthened as follows.

Corollary 1 (Softmax regularity with improved constants). *For any $\lambda > 0$, the softmax map $\sigma_\lambda : \mathbb{R}^n \rightarrow \Delta_n^\circ$ satisfies*

$$\begin{aligned} (\text{Lipschitz}) \quad & \|\sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y})\|_p \leq \frac{\lambda}{2} \|\mathbf{x} - \mathbf{y}\|_p, \quad \forall p \geq 1, \\ (\text{co-coercive}) \quad & \langle \sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq \frac{2}{\lambda} \|\sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y})\|_2^2. \end{aligned}$$

In particular,

- σ_λ is nonexpansive and firmly nonexpansive for $\lambda \in (0, 2]$;
- σ_λ is contractive for $\lambda \in (0, 2)$.

The proof of this corollary follows directly from the analysis in Gao & Pavel (2017), with the Lipschitz constant of the softmax operator replaced by the sharper value $\lambda/2$.

4.2 Lipschitz Analysis of Attention variant

The self-attention module (Vaswani et al., 2017) refines an input feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ by projecting it into queries, keys, and values via learnable matrices $\mathbf{W}^Q \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}^K \in \mathbb{R}^{d \times d_k}$, and $\mathbf{W}^V \in \mathbb{R}^{d \times d_v}$. The attention operation is then given by

$$\text{Att}(\mathbf{X}; \mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V) = \text{softmax}\left(\frac{\mathbf{X}\mathbf{W}^Q(\mathbf{X}\mathbf{W}^K)^\top}{\sqrt{d_k}}\right)\mathbf{X}\mathbf{W}^V, \quad (5)$$

where $\text{softmax}(\cdot)$ denotes the row-wise application of the function σ_1 . Although we obtain a tight global Lipschitz constant for the softmax operator σ_1 acting on the rows of the attention score matrix, the self-attention map $\mathbf{X} \mapsto \text{Att}(\mathbf{X}; \mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V)$ defined on an unbounded input domain is not globally Lipschitz. Kim et al. (Kim et al., 2021) show that, for this map, the Lipschitz constant with respect to any ℓ_p norm is infinite. Motivated by this negative result, Qi et al. (2023) propose *scaled cosine similarity attention* (SCSA) as a Lipschitz-controlled variant of self-attention, which normalizes the projections $\mathbf{X}\mathbf{W}^Q$ and $\mathbf{X}\mathbf{W}^K$ to unit norm and applies an inverse temperature coefficient $\tau > 0$:

$$\text{SCSA}(\mathbf{X}; \mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V, \nu, \tau) = \nu \text{softmax}\left(\tau \frac{\mathbf{X}\mathbf{W}^Q(\mathbf{X}\mathbf{W}^K)^\top}{\|\mathbf{X}\mathbf{W}^Q\| \|\mathbf{X}\mathbf{W}^K\|}\right)\mathbf{W}^V. \quad (6)$$

SCSA can be viewed as a Lipschitz variant of the standard attention mechanism. Theorem 1 in Qi et al. (2023) establishes an Lipschitz bound for scaled cosine similarity attention (SCSA), but their bound depends on the choice $(n-1)/n$ as a Lipschitz constant for the softmax operator. Using our sharper analysis of the softmax Lipschitz constant, we can replace $(n-1)/n$ by $1/2$ in their argument. Thus Lipschitz bound for SCSA can be tightened as shown below.

Theorem 2 (Refined ℓ_2 -Lipschitz bound for SCSA). *Let $L_2(\text{SCSA})$ denote the global Lipschitz constant of $\mathbf{X} \mapsto \text{SCSA}(\mathbf{X}, \mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V, \nu, \tau)$ with respect to the ℓ_2 norm. Then, for fixed $(\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V, \nu, \tau)$,*

$$L_2(\text{SCSA}) \leq n^2 \nu \tau \varepsilon^{-1/2} \|\mathbf{W}^K\|_2 + n \nu \tau \varepsilon^{-1/2} \|\mathbf{W}^Q\|_2 + 2n \nu \varepsilon^{-1/2} \|\mathbf{W}^{V^\top}\|_2.$$

4 改进后的 Softmax 利普希茨常数的影响

定理1中的结果表明，softmax函数对于所有 ℓ_p 范数在 $p \geq 1$ 上是 $\lambda/2$ -利普希茨的。这种更紧密的表征使我们能够重新审视现有结果，其中softmax利普希茨连续性是相关的，从而得到更精确的常数或简化的等式。接下来，我们在几个代表性设置中说明这一点。

4.1 高 & 佩夫尔（2017年，引理3）的改进

利用我们更精确的利普希茨估计，高 & 佩夫尔（2017年）中的引理3可以加强如下。**引理1** (Softmax 正则化与改进常数). 对于任何 $\lambda > 0$, *Softmax* 映射 $\sigma_\lambda : \mathbb{R}^n \rightarrow \Delta_n^\circ$ 满足

$$\begin{aligned} (\text{Lipschitz}) \quad & \|\sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y})\|_p \leq \frac{\lambda}{2} \|\mathbf{x} - \mathbf{y}\|_p, \quad \forall p \geq 1, \\ (\text{co-coercive}) \quad & \langle \sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq \frac{2}{\lambda} \|\sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y})\|_2^2. \end{aligned}$$

特别是,

- σ_λ 对于 $\lambda \in (0, 2]$ 是非扩张的且是强非扩张的
- σ_λ 在 $\lambda \in (0, 2)$ 上是收缩映射。

这个推论的证明直接来自Gao & Pavel (2017)的分析，其中softmax算子的利普希茨常数被替换为更精确的值 $\lambda/2$ 。

4.2 注意力变体的利普希茨分析

自注意力模块（Vaswani等人，2017）通过可学习的矩阵 $\mathbf{W}^Q \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}^K \in \mathbb{R}^{d \times d_k}$ 和 $\mathbf{W}^V \in \mathbb{R}^{d \times d_v}$ 将输入特征矩阵 $\mathbf{X} \in \mathbb{R}^{n \times d}$ 投影到查询、键和值中。然后注意力操作由

$$\text{Att}(\mathbf{X}; \mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V) = \text{softmax}\left(\frac{\mathbf{X}\mathbf{W}^Q(\mathbf{X}\mathbf{W}^K)^\top}{\sqrt{d_k}}\right)\mathbf{X}\mathbf{W}^V, \quad (5)$$

其中 $\text{softmax}(\cdot)$ 表示函数 σ_1 的按行应用。尽管我们为作用于注意力得分矩阵行的softmax算子 σ_1 获得了一个紧密的全局利普希茨常数，但在无界输入域上定义自注意力映射 $\mathbf{X} \mapsto \text{Att}(\mathbf{X}; \mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V)$ 不是全局利普希茨的。Kim等人（Kim等人，2021）表明，对于这个映射，相对于任何 ℓ_p 范数的利普希茨常数是无限的。受此负面结果的启发，Qi等人（2023）提出了缩放余弦相似度注意力(SCSA)作为利普希茨控制的注意力变体，它将投影 $\mathbf{X}\mathbf{W}^Q$ 和 $\mathbf{X}\mathbf{W}^K$ 归一化到单位范数并应用逆温度系数 $\tau > 0$:

$$\text{SCSA}(\mathbf{X}; \mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V, \nu, \tau) = \nu \text{softmax}\left(\tau \frac{\mathbf{X}\mathbf{W}^Q(\mathbf{X}\mathbf{W}^K)^\top}{\|\mathbf{X}\mathbf{W}^Q\| \|\mathbf{X}\mathbf{W}^K\|}\right)\mathbf{W}^V. \quad (6)$$

SCSA可以被视为标准注意力机制的一种利普希茨变体。Qi等人（2023年）的定理1为缩放余弦相似度注意力（SCSA）建立了利普希茨界，但他们的界取决于将 $(n-1)/n$ 作为softmax算子的利普希茨常数的选择。利用我们对softmax利普希茨常数的更精确分析，我们可以在他们的论证中将 $(n-1)/n$ 替换为 $1/2$ 。因此，SCSA的利普希茨界可以被进一步收紧，如下所示。

定理2 (改进的 ℓ_2 -利普希茨界 for SCSA). 令 $L_2(\text{SCSA})$ 表示 $\text{SCSA}(\mathbf{X}, \mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V, \nu, \tau)$ 关于 ℓ_2 范数的全局利普希茨常数。那么，对于固定的 $(\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V, \nu, \tau)$,

$$L_2(\text{SCSA}) \leq n^2 \nu \tau \varepsilon^{-1/2} \|\mathbf{W}^K\|_2 + n \nu \tau \varepsilon^{-1/2} \|\mathbf{W}^Q\|_2 + 2n \nu \varepsilon^{-1/2} \|\mathbf{W}^{V^\top}\|_2.$$

In particular, compared with the Lipschitz bound stated in Theorem 1 of Qi et al. (2023), Theorem 2 improves the estimate by a factor of 2, removing the extra factor 2 that appears in their original bound. Such refinements can directly impact theoretical robustness guarantees for attention-based architectures.

4.3 Entropy-regularized zero-sum games via double-softmax fixed point

Consider a two-player zero-sum matrix game with a payoff matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$. The standard minimax formulation is

$$\min_{\mathbf{x} \in \Delta_n} \max_{\mathbf{y} \in \Delta_m} \mathbf{x}^\top \mathbf{A} \mathbf{y},$$

where Δ_n, Δ_m denote the probability simplices. This formulation characterizes the Nash equilibrium of the game, capturing the row player’s strategy that minimizes the worst-case loss against an adversarial opponent, and thus serves as a foundation for robust decision making, reinforcement learning, and adversarial training. To ensure unique mixed strategies and to improve algorithmic stability, it is common to add entropic regularization terms (Cen et al., 2021). The entropy-regularized problem becomes

$$\min_{\mathbf{x} \in \Delta_n} \max_{\mathbf{y} \in \Delta_m} \mathbf{x}^\top \mathbf{A} \mathbf{y} + \tau (H(\mathbf{x}) - H(\mathbf{y})),$$

where, $\tau > 0$ is the regularization parameter and $H(\cdot)$ is the Shannon entropy, defined for any probability vector $\mathbf{x} \in \Delta_n$ as $H(\mathbf{x}) = -\sum_{i=1}^n x_i \log x_i$.

A well-known algorithm to solve the above entropy regularization problem is the double-softmax fixed-point (DSFP) iteration (McKelvey & Palfrey, 1995), which updates

$$\mathbf{y}_{k+1} \leftarrow (1 - \alpha) \mathbf{y}_k + \alpha \sigma_{1/\tau}(\mathbf{A}^\top \sigma_{1/\tau}(-\mathbf{A} \mathbf{y}_k)).$$

Intuitively, each player responds via a softmax update, yielding an equilibrium. Our improved Lipschitz constant for softmax in Theorem 1 provides a tighter condition to choose the regularization parameter τ depending on the payoff matrix $\mathbf{A}$, such that the DSFP iteration is contractive and convergent.

Theorem 3 (Convergence of DSFP under entropy regularization). *Let $\mathbf{A} \in \mathbb{R}^{n \times m}$ be a payoff matrix and let $\tau > 0$ denote the entropy regularization parameter. For all $1 \leq p \leq \infty$, if $\tau > \|\mathbf{A}\|_p/2$ and $\alpha \in (0, 1]$, then the DSFP iteration is a Banach contraction on $(\Delta_m, \|\cdot\|_p)$ with contraction factor $\|\mathbf{A}\|_p^2/(4\tau^2)$. Hence, the iterates $\mathbf{y}_k$ converge linearly in the ℓ_p norm to the unique fixed point $\mathbf{y}^* \in \Delta_m$. The corresponding strategy of the row player is recovered as*

$$\mathbf{x}^* = \sigma_{1/\tau}(-\mathbf{A} \mathbf{y}^*),$$

which is also convergent since $\mathbf{x}_k = \sigma_{1/\tau}(-\mathbf{A} \mathbf{y}_k)$ at each step.

In summary, for DSFP iterations to converge linearly under the ℓ_p norm, we can derive a condition for the regularization parameter τ to satisfy and thereby guarantee well-defined equilibrium strategies.

5 Empirical Validation of Lipschitz Constant

In this section, we empirically validate our theoretical result that the softmax operator has a global Lipschitz constant of 1/2, which is norm-uniform and tight. Our experiments involve a broad range of architectures across vision, language, and reinforcement learning, under varying datasets, prompts, and inverse temperature coefficient λ . All experiments were conducted on NVIDIA A40 GPUs. We compute the empirical Lipschitz constant using the definition in equation 3, using pairs of input obtained from each experimental setting. In particular, we compute,

$$\text{Empirical } L_p = \max_{1 \leq i \leq M} \frac{\|\sigma_\lambda(\mathbf{x}_i) - \sigma_\lambda(\mathbf{x}_i + \delta \mathbf{x}_i)\|_p}{\|\delta \mathbf{x}_i\|_p}, \quad (7)$$

where M is the number of inputs. For each input sample $\mathbf{x}_i$, we apply small random perturbations $\delta \mathbf{x}_i$ normalized to have $\|\delta \mathbf{x}_i\|_p = \epsilon$, where ϵ denotes the perturbation magnitude. We report the computed Empirical L_p constant across multiple values of ϵ and p .

特别地，与Qi等人(2023)在定理1中提出的利普希茨界相比，定理2将估计值提高了2倍，消除了他们原始界中出现的额外因子2。此类改进可以直接影响基于注意力架构的理论鲁棒性保证。

4.3 基于双重softmax固定点的熵正则化零和博弈

考虑一个具有支付矩阵 $\mathbf{A} \in \mathbb{R}^{n \times m}$ 的两玩家零和矩阵博弈。标准的minimax公式为

$$\min_{\mathbf{x} \in \Delta_n} \max_{\mathbf{y} \in \Delta_m} \mathbf{x}^\top \mathbf{A} \mathbf{y},$$

其中 Δ_n, Δ_m 表示概率单纯形。该公式刻画了博弈的纳什均衡，捕捉了行玩家在对抗性对手面前的最坏情况损失最小化的策略，因此为鲁棒决策、强化学习和对抗训练奠定了基础。为确保混合策略的唯一性和提高算法稳定性，通常添加熵正则化项（Cen等人，2021年）。熵正则化问题变为

$$\min_{\mathbf{x} \in \Delta_n} \max_{\mathbf{y} \in \Delta_m} \mathbf{x}^\top \mathbf{A} \mathbf{y} + \tau (H(\mathbf{x}) - H(\mathbf{y})),$$

其中， $\tau > 0$ 是正则化参数， $H(\cdot)$ 是香农熵，对于任何概率向量 $\mathbf{x} \in \Delta_n$ 定义为 $H(\mathbf{x}) = -\sum_{i=1}^n x_i \log x_i$

一种众所周知的算法，用于解决上述熵正则化问题，是双softmax固定点（DSFP）迭代（McKelvey & Palfrey, 1995），该算法更新

$$\mathbf{y}_{k+1} \leftarrow (1 - \alpha) \mathbf{y}_k + \alpha \sigma_{1/\tau}(\mathbf{A}^\top \sigma_{1/\tau}(-\mathbf{A} \mathbf{y}_k)).$$

直观地讲，每个玩家通过softmax更新做出响应，从而得到一个均衡。定理1中softmax的利普希茨常数为我们提供了一种更紧密的条件来选择正则化参数 τ ，该参数取决于支付矩阵 $\mathbf{A}$ ，使得DSFP迭代是收缩映射且收敛的。

定理3 (熵正则化下的DSFP收敛性)。设 $\mathbf{A} \in \mathbb{R}^{n \times m}$ 为支付矩阵，设 $\tau > 0$ 表示熵正则化参数。对于所有 $1 \leq p \leq \infty$ ，如果 $\tau > \|\mathbf{A}\|_p/2$ 且 $\alpha \in (0, 1]$ ，则 **DSFP** 迭代在 $(\Delta_m, \|\cdot\|_p)$ 上是一个 **Banach** 收缩映射，收缩因子为 $\|\mathbf{A}\|_p^2/(4\tau^2)$ 。因此，迭代序列 $\mathbf{y}_k$ 在 ℓ_p 范数下线性收敛到唯一的固定点 $\mathbf{y}^* \in \Delta_m$ 。相应的行玩家的策略被恢复为

$$\mathbf{x}^* = \sigma_{1/\tau}(-\mathbf{A} \mathbf{y}^*),$$

这也是收敛的，因为在每一步中， $\mathbf{x}_k = \sigma_{1/\tau}(-\mathbf{A} \mathbf{y}_k)$ 。

总之，为了使DSFP迭代在 ℓ_p 范数下线性收敛，我们可以推导出正则化参数 τ 需要满足的条件，从而保证均衡策略的有效定义。

5 经验验证利普希茨常数

在本节中，我们通过实验验证了我们的理论结果：softmax算子具有全局Lipschitz常数 1/2，该常数是范数一致的且是紧的。我们的实验涵盖了视觉、语言和强化学习等多个领域的广泛架构，在变化的数据集、提示和逆温度系数 λ 下进行。所有实验均在NVIDIA A40 GPU上完成。我们使用方程3中的定义计算经验利普希茨常数，使用从每个实验设置中获得的输入对进行计算。特别是，我们计算，

$$\text{Empirical } L_p = \max_{1 \leq i \leq M} \frac{\|\sigma_\lambda(\mathbf{x}_i) - \sigma_\lambda(\mathbf{x}_i + \delta \mathbf{x}_i)\|_p}{\|\delta \mathbf{x}_i\|_p}, \quad (7)$$

M 是输入的数量。对于每个输入样本 $\mathbf{x}_i$ ，我们应用小的随机扰动 $\delta \mathbf{x}_i$ ，这些扰动被归一化以具有 $\|\delta \mathbf{x}_i\|_p = \epsilon$ ，其中 ϵ 表示扰动幅度。我们报告了在多个 ϵ 和 p 值下计算的经验 L_p 常数。

5.1 Vision Models

We empirically evaluate the Lipschitz constant of the softmax operator, which appears in the attention mechanism of vision models. In particular, we consider Vision Transformer (ViT) models (Dosovitskiy et al., 2021) in three variants, Base (86M parameters), Large (307M parameters), and Huge (632M parameters). For each model, we select images from both the CIFAR-100 and ImageNet datasets, pass them through the transformer, and extract the pre-softmax attention scores $QK^\top/\sqrt{d_k}$, which is the input to the softmax in Eq. 5. We then apply perturbations to these scores and compute the empirical Lipschitz constant row-wise according to Definition 7. M is set to be 100 images.

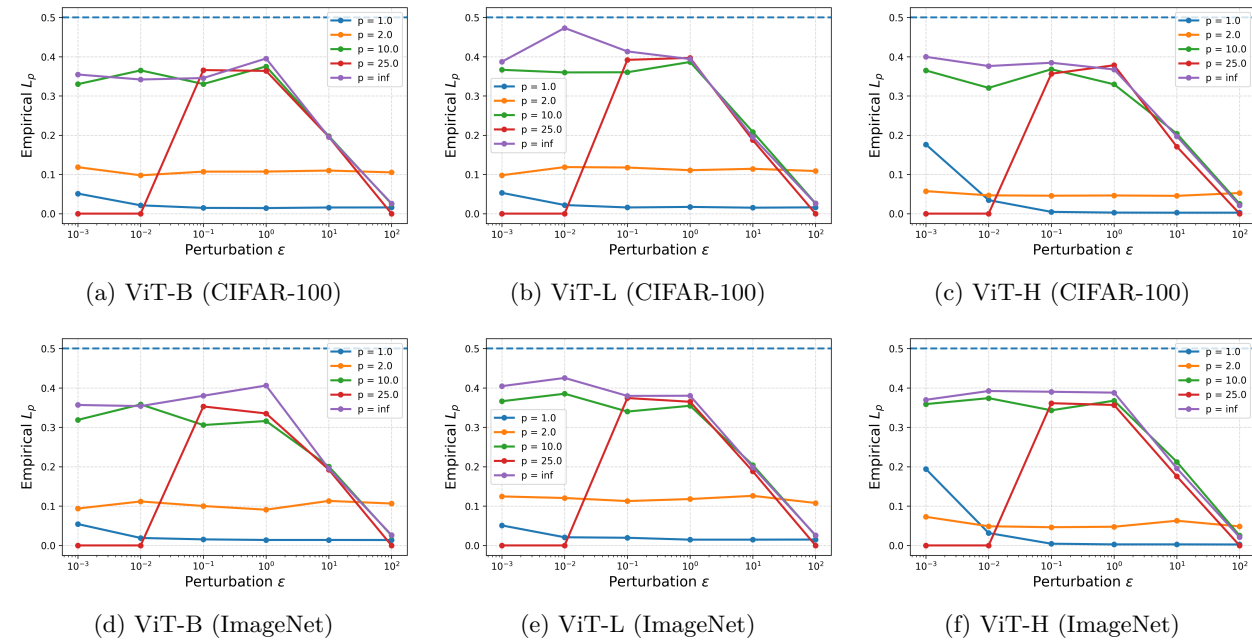


Figure 1: Empirical L_p of the softmax operator over attention scores from three Vision Transformer (ViT) variants on CIFAR-100 (top row) and ImageNet (bottom row), across varying perturbation magnitudes ϵ for multiple ℓ_p norms. In all cases, the empirical values remain below the derived bound of $1/2$.

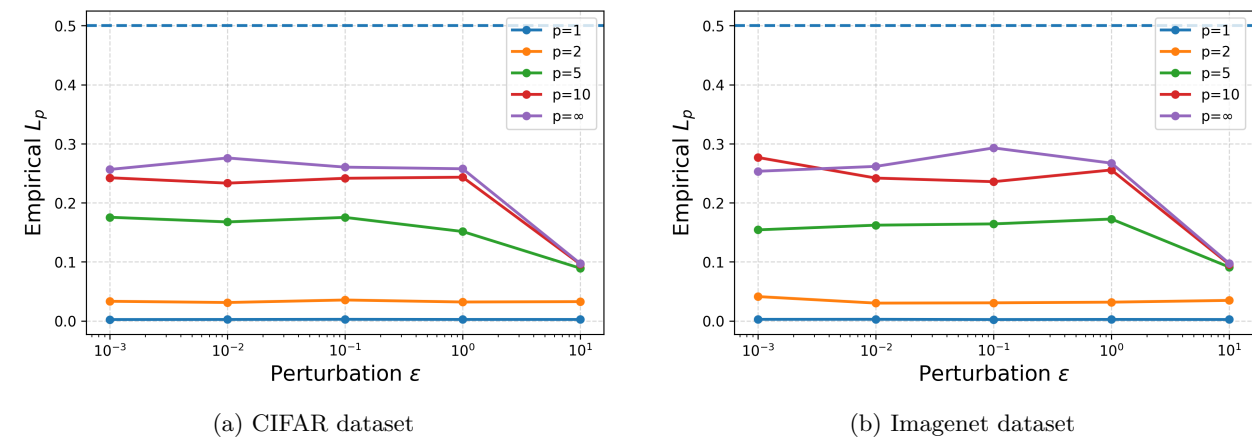


Figure 2: Empirical L_p of the softmax operator for classification logits of ResNET50 model on (a) CIFAR-100 and (b) ImageNet dataset, across varying perturbation magnitudes ϵ for multiple ℓ_p norms. In all cases, the empirical values remain well below the derived bound of $1/2$.

5.1 视觉模型

我们通过经验评估softmax算子的利普希茨常数，该常数出现在视觉模型的注意力机制中。特别是，我们考虑了Dosovitskiy等人（2021年）提出的视觉Transformer (ViT) 模型的三种变体：基础版（86M参数）、大型版（307M参数）和巨型版（632M参数）。对于每个模型，我们从CIFAR-100和ImageNet数据集中选择图像，将它们通过transformer，并提取预softmax注意力分数 $QK^\top/\sqrt{d_k}$ ，这是Eq. 5中softmax的输入。然后我们对这些分数应用扰动，并根据定义7按行计算经验利普希茨常数。 M 设置为100张图像。

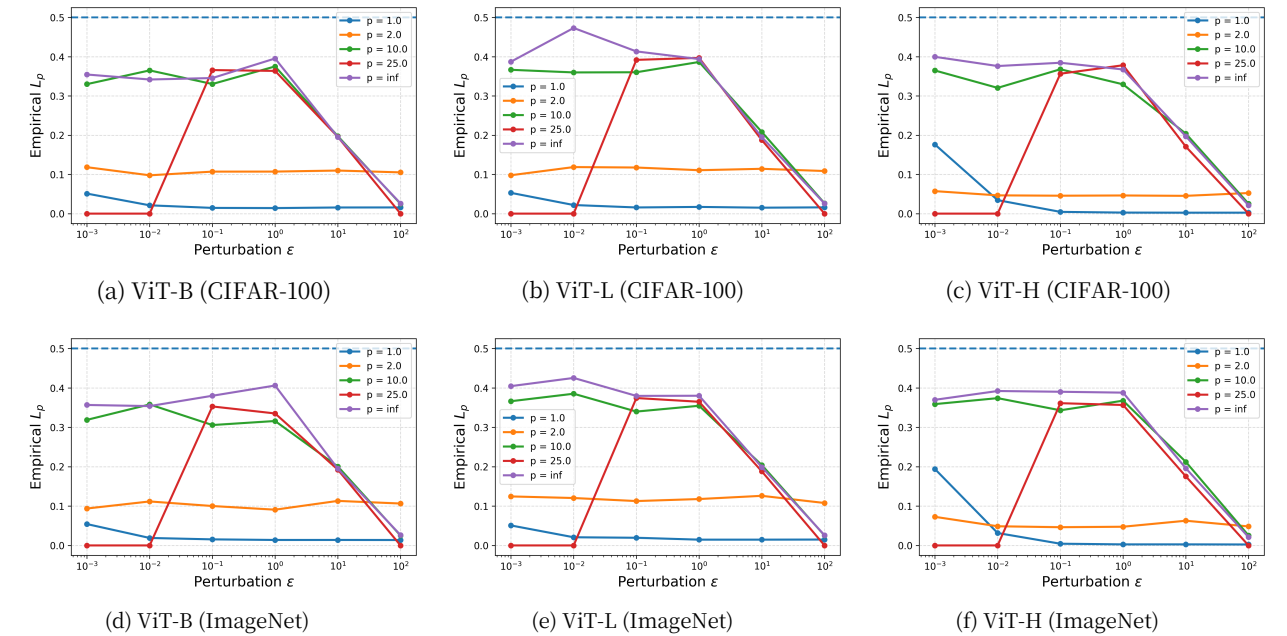


图1: softmax算子在不同扰动幅度 ϵ 下，对三个视觉Transformer (ViT) 变体在CIFAR-100（第一行）和ImageNet（第二行）上注意力分数的实验 L_p 。所有情况下，实验值均低于推导出的界限 $1/2$ 。

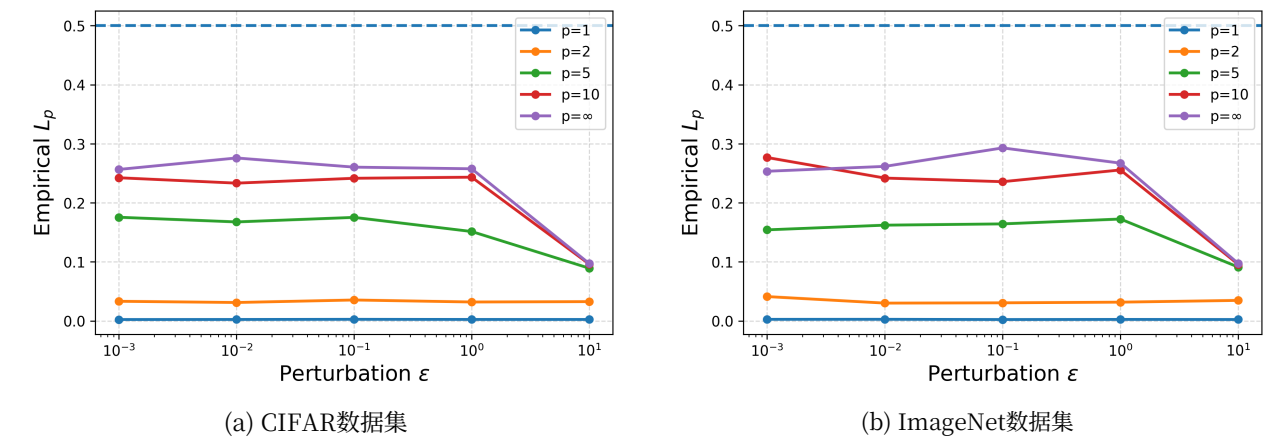


图2: ResNET50模型在不同扰动幅度 ϵ 下，对分类logits在(a) CIFAR-100和(b) ImageNet数据集上的softmax算子的实验 L_p 。所有情况下，实验值均远低于推导出的界限 $1/2$ 。

The results are presented as plots of empirical Lipschitz values versus the perturbation norm in Fig. 1 for CIFAR-100 (top row) and ImageNet (bottom row) datasets, with the theoretical baseline of 0.5 indicated. We report results across multiple ℓ_p norms. In each case, the plotted value corresponds to the maximum empirical Lipschitz constant over all attention heads, layers, tokens, and images. In particular, the total number of tested vectors is significant, which is equal to $N \times H \times M \times 12$, where N is the number of tokens and H the number of heads. The plots consistently show that the empirical Lipschitz constant remains below our theoretical value of $1/2$, regardless of the choice of norm or ViT variant.

We further evaluate the Lipschitz constant of the softmax operator on the classification logits of ResNET50 model (He et al., 2016), using images from the CIFAR-100 dataset in Fig. 2a and ImageNet dataset in Fig. 2b. In this experiment, we use $M = 25,000$ images and introduce input perturbations to the logits and compute the empirical L_p constant, for different p and perturbation magnitude ϵ . Consistent with our findings for attention scores, the empirical L_p remains strictly below the theoretical bound of $1/2$.

5.2 Language Models

We next evaluate the Lipschitz constant of the softmax operator on attention scores of large language models. Specifically, we sample 100 prompts each from the HellaSwag (Zellers et al., 2019) and the PIQA (Bisk et al., 2020) datasets, process them through GPT-2 (Radford et al., 2019) (518M parameters) and Qwen3 (Bai et al., 2023) (8B parameters), extract the pre-softmax attention scores, apply perturbations, and compute the empirical Lipschitz constant as defined in Definition 7.

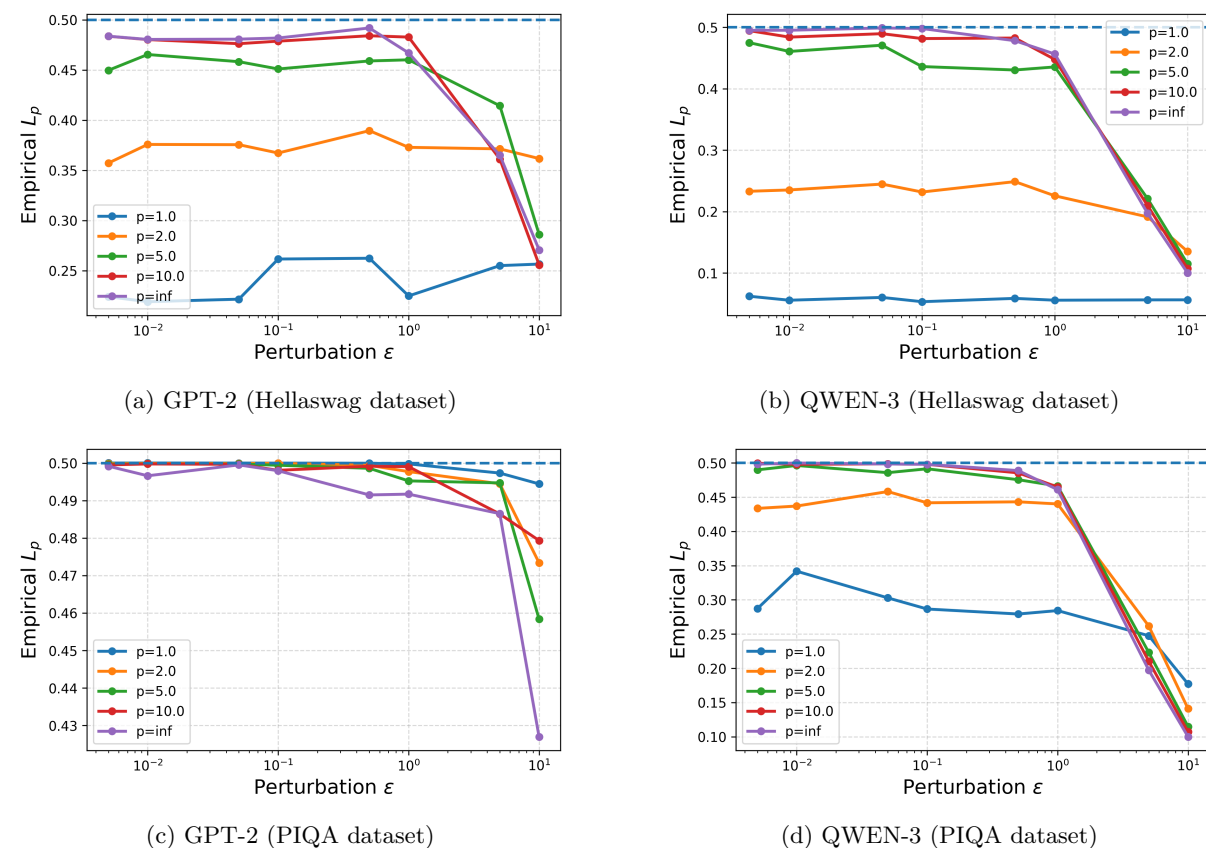


Figure 3: Empirical L_p of the softmax operator over attention scores from two Large Language models, GPT-2 and QWEN-3, on HellaSwag dataset (top row) and PIQA dataset (bottom row), across varying perturbation magnitudes ϵ for multiple ℓ_p norms. Across all configurations, the empirical values remain below the theoretical bound of $1/2$, with several instances approaching this limit, thereby confirming the tightness of the derived bound.

结果以图1中经验利普希茨值与扰动范数的图像形式呈现，分别针对CIFAR-100（顶行）和ImageNet（底行）数据集，其中理论基线为0.5。我们报告了跨多种 ℓ_p 范数的结果。每种情况下，图中所示的值对应于所有注意力头、层、token和图像中的最大经验利普希茨常数。特别是，测试向量的总数非常显著，等于 $N \times H \times M \times 12$ ，其中 N 是token数量， H 是注意力头数量。图像始终显示，无论选择何种范数或ViT变体，经验利普希茨常数都保持在我们的理论值 $1/2$ 以下。

我们进一步评估了ResNET50模型（He et al., 2016）分类logits上softmax算子的利普希茨常数，使用CIFAR-100数据集（图2a）和ImageNet数据集（图2b）中的图像。在此实验中，我们使用 $M = 25,000$ 张图像，对logits引入输入扰动，并计算不同 p 和扰动幅度 ϵ 的经验 L_p 常数。与我们对注意力分数的发现一致，经验 L_p 严格低于理论界限 $1/2$ 。

5.2 语言模型

接下来，我们评估了softmax算子对大型语言模型注意力分数的利普希茨常数。具体来说，我们从HellaSwag（Zellers等人，2019年）和PIQA（Bisk等人，2020年）数据集分别采样100个提示，通过GPT-2（Radford等人，2019年）（518M参数）和Qwen3（Bai等人，2023年）（8B参数）进行处理，提取预softmax注意力分数，施加扰动，并计算定义7中定义的经验利普希茨常数。

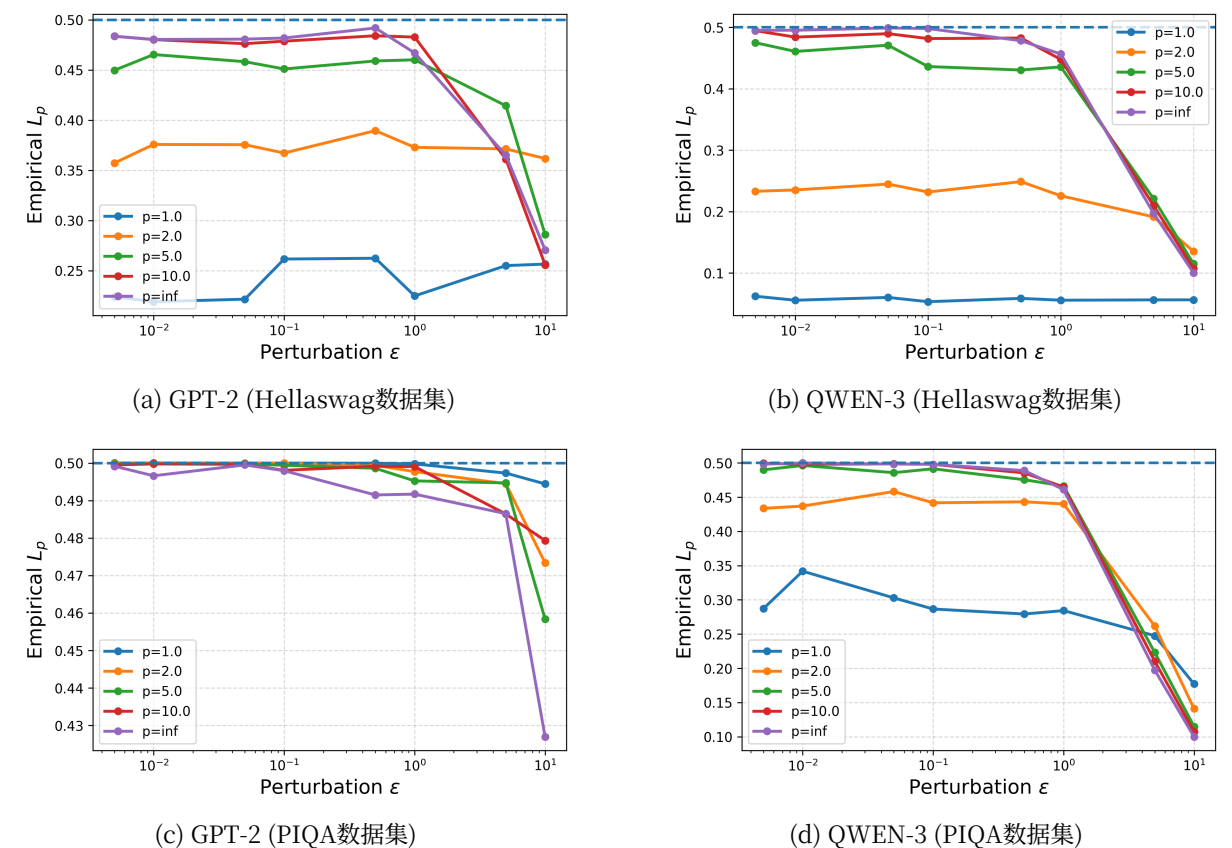


图3: softmax算子在不同扰动幅度 ϵ 下，对两个大型语言模型GPT-2和QWEN-3在HellaSwag数据集（顶行）和PIQA数据集（底行）上的注意力分数的实验 L_p 。在所有配置下，实验值均低于理论界限 $1/2$ ，部分实例接近该极限，从而验证了所推导界限的紧致性。

The scale of this evaluation is substantial here as well. The total number of tested vectors equals $H \times 100 \times 12$ per token per prompt, where the number of tokens varies with prompt length. Consistent with our experiments on vision models, the results in Fig. 3 for the Hellaswag dataset (top row) and the PIQA dataset (bottom row) demonstrate that the empirical Lipschitz constant remains strictly below the theoretical value of $1/2$ across all norms. Notably, for the PIQA dataset, we observe empirical constants of 0.4995 for the Qwen3 model and 0.4999 for GPT-2, providing empirical evidence for the tightness of our theoretically derived Lipschitz bound.

5.3 Reinforcement Learning Policies

We next investigate the Lipschitz behavior of softmax within stochastic policies in reinforcement learning. In policy-gradient methods (Sutton & Barto, 2018), a stochastic policy is typically parameterized as

$$\pi_\lambda(a | s) = \frac{\exp(\lambda Q(s, a))}{\sum_{a' \in \mathcal{A}} \exp(\lambda Q(s, a'))},$$

where $Q(s, a)$ are the action logits for state $s \in \mathcal{S}$, $\mathcal{A}$ is the action space, and $\lambda > 0$ is the inverse temperature coefficient. We evaluate this mapping $Q(s, \cdot) \mapsto \pi_\lambda(\cdot | s)$ in the **LunarLander** environment (discrete action space of size $|\mathcal{A}| = 4$) and the **CartPole** environment ($|\mathcal{A}| = 2$) from the OpenAI Gym benchmark (Brockman et al., 2016). We implement a PPO (Proximal Policy Optimization) agent (Schulman et al., 2017) using the **stable_baselines3** library, initialized randomly, and executed to collect a large set of visited states. For each state, we perturb the corresponding action logits $Q(s, \cdot)$ with random perturbations, and compute the empirical Lipschitz constant as in Definition 7. The reported value is obtained by averaging over multiple states ($M = 25000$) and perturbation trials for different coefficients λ . The results in Fig. 4 for Cartpole and Lunarlander environments confirm that the empirical constants scale proportionally to $\lambda/2$, in agreement with our theoretical prediction. Note that for the **CartPole** environment, where the softmax input dimension is $n = 2$, we obtain an empirical constant of 1.9996 for $p = 8$ and $\lambda = 4.0$, which is very close to the derived bound.

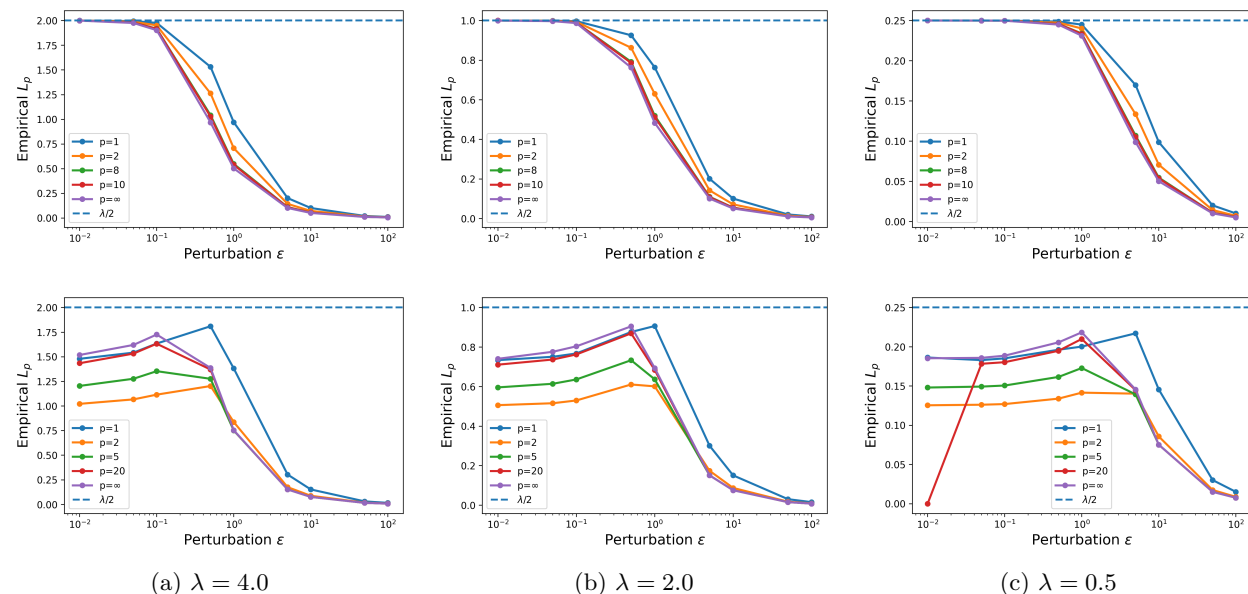


Figure 4: Empirical Lipschitz sensitivity of the softmax policy in RL environments, Cartpole (top row) and Lunarlander (bottom row), across varying perturbation magnitudes ϵ , under varying coefficients λ for different p -norms. The empirical values scale with λ and remain below the theoretical bound of $\lambda/2$, thereby empirically confirming the derived bound.

In summary, across all models, datasets, and settings considered, the empirical Lipschitz constant of softmax never exceeded $1/2$, thereby providing strong empirical validation of our theoretical bound. Furthermore, in

这次评估的规模也相当大。测试向量的总数等于 $H \times 100 \times 12$ 每个token每个提示，其中token数量随提示长度变化。与我们在视觉模型上的实验结果一致，图3中Hellaswag数据集（顶行）和PIQA数据集（底行）的结果表明，经验利普希茨常数在所有范数下都严格低于理论值 $1/2$ 。值得注意的是，对于PIQA数据集，我们观察到Qwen3模型的经验常数为0.4995，GPT-2的经验常数为0.4999，为我们的理论推导出的利普希茨界提供了经验证据。

5.3 强化学习策略

我们接下来研究强化学习中随机策略下softmax的利普希茨行为。在策略梯度方法（Sutton & Barto, 2018）中，随机策略通常参数化为

$$\pi_\lambda(a | s) = \frac{\exp(\lambda Q(s, a))}{\sum_{a' \in \mathcal{A}} \exp(\lambda Q(s, a'))},$$

其中 $Q(s, a)$ 是状态 $s \in \mathcal{S}$ 的动作logits， $\mathcal{A}$ 是动作空间， $\lambda > 0$ 是逆温度系数。我们在 **LunarLander** 环境（离散动作空间大小为 $|\mathcal{A}| = 4$ ）和 **CartPole** 环境（ $|\mathcal{A}| = 2$ ）中从OpenAI Gym基准（Brockman et al., 2016）评估这个映射 $Q(s, \cdot) \mapsto \pi_\lambda(\cdot | s)$ 。我们使用 **stable_baselines3** 库实现PPO（近端策略优化）代理（Schulman et al., 2017），随机初始化，并执行以收集大量访问过的状态。对于每个状态，我们对相应的动作logits $Q(s, \cdot)$ 进行随机扰动，并按照定义7计算经验利普希茨常数。报告的值是通过在多个状态 ($M = 25000$) 和不同系数 λ 的扰动试验上取平均值获得的。图4中Cartpole和Lunarlander环境的实验结果证实，经验常数与 $\lambda/2$ 成比例，与我们的理论预测一致。注意对于 **CartPole** 环境，其中softmax输入维度为 $n = 2$ ，我们为 $p = 8$ 和 $\lambda = 4.0$ 得到经验常数为 1.9996，这非常接近推导出的界限。

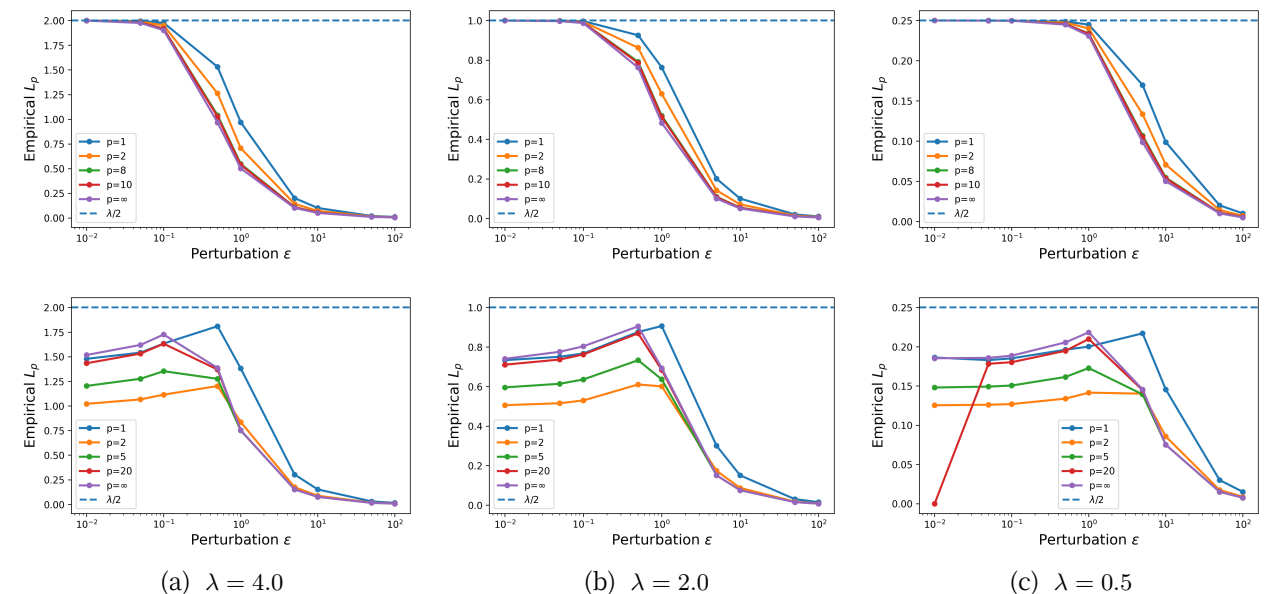


图4：强化学习环境中softmax策略的经验Lipschitz敏感性，Cartpole（第一行）和Lunarlander（第二行），在不同扰动幅度 ϵ 下，以及针对不同 p -范数的不同系数 λ 。经验值与 λ 成正比，并保持在理论界限 $\lambda/2$ 之下，从而经验上验证了所推导的界限。

总而言之，在所有考虑的模型、数据集和设置中，softmax的经验利普希茨常数从未超过 $1/2$ ，从而为我们的理论界限提供了强有力的经验验证。此外，在

several cases, the empirical estimates approached $1/2$, illustrating that the derived bound is indeed tight in practice.

Conclusion

In this work, we derived that the softmax function admits a Lipschitz constant of $1/2$ uniformly across all ℓ_p norms. This result clarifies a commonly adopted assumption in the machine learning literature, where the Lipschitz constant of the softmax operator has often been considered as 1 with respect to the ℓ_2 norm. We further prove the tightness of our derived bound and demonstrate its utility in strengthening existing theoretical analyses. We also estimate the empirical Lipschitz constants for softmax transformations applied to attention score matrices from transformer-based architectures such as ViT, GPT-2, and Qwen3, classification logits from ResNet-50, and stochastic policies in reinforcement learning agents. In all cases, the empirical estimates remain strictly below the theoretical bound of $1/2$, thereby guaranteeing the tightness and generality of our results.

Broader Impact Statement

This work is primarily theoretical. By itself, this result does not introduce new datasets, architectures, or applications, but it provides sharper analytical tools for reasoning about the stability and robustness of existing models. On the positive side, such tools can help practitioners design systems with better controlled sensitivity to perturbations, complementing existing Lipschitz-based approaches used to improve robustness and training stability in deep networks (Cisse et al., 2017)(Miyato et al., 2018). At the same time, any advance that makes robust and stable deployment easier can be dual-use - it may also strengthen models used in societally sensitive settings, including surveillance and biometric monitoring (Almeida et al., 2022). A further risk is overstatement or misinterpretation - guarantees for a single component can be misconstrued as end-to-end assurances of safety, fairness, or reliability, which could encourage premature use in high-stakes contexts (AI, 2023).

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几种情况下, 经验估计值接近 $1/2$, 说明所得到的界限在实践中确实是紧的。

结论

在本工作中, 我们推导出softmax函数在所有 ℓ_p 范数下具有 $1/2$ 的利普希茨常数。这一结果澄清了机器学习文献中普遍采用的一个假设, 即softmax算子的利普希茨常数相对于 ℓ_2 范数通常被认为是1。我们进一步证明了所推导边界的紧致性, 并展示了其在加强现有理论分析方面的效用。我们还估计了应用于ViT、GPT-2和Qwen3等基于Transformer架构的注意力得分矩阵、ResNet-50的分类logits以及强化学习智能体中的随机策略的softmax变换的经验利普希茨常数。在所有情况下, 经验估计值均严格低于理论边界 $1/2$, 从而保证了我们结果的紧致性和普适性。

更广泛的影响声明

这项工作主要是理论性的。仅凭这一结果本身并未引入新的数据集、架构或应用, 但它为推理现有模型的稳定性和鲁棒性提供了更精确的分析工具。从积极方面来看, 此类工具可以帮助从业者设计对抗动具有更好控制敏感性的系统, 补充现有的基于利普希茨的方法, 这些方法用于提高深度网络的鲁棒性和训练稳定性 (Cisse等人, 2017) (Miyato等人, 2018)。同时, 任何使鲁棒和稳定部署更容易的进展都可能具有双重用途——它也可能加强在社会敏感环境中使用的模型, 包括监控和生物特征监测 (Almeida等人, 2022)。进一步的风险是夸大或误读——单个组件的保证可能被误解为端到端的、安全性、公平性或可靠性的保证, 这可能会鼓励在高风险背景下过早使用 (AI, 2023)。

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A Appendix

A.1 Proof of Proposition 1

(a) and (b) follow from standard results in matrix analysis (Golub & Van Loan, 2013).

Proof of Proposition 1(c). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. For $p = 1$ and $p = \infty$, the identity trivially satisfies, and hence fix $1 < p < \infty$. For any $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{Ax}\|_p^p = \sum_{i=1}^n |(\mathbf{Ax})_i|^p,$$

where $(\mathbf{Ax})_i$ denotes i^{th} element in $\mathbf{Ax}$. Then, for each i , the triangle inequality gives

$$|(\mathbf{Ax})_i| = \left| \sum_{j=1}^n A_{ij}x_j \right| \leq \sum_{j=1}^n |A_{ij}| |x_j|$$

By Holder's inequality, we obtain for each i ,

$$\sum_{j=1}^n |A_{ij}| |x_j| \leq \left(\sum_{j=1}^n |A_{ij}| \right)^{1-1/p} \left(\sum_{j=1}^n |A_{ij}| |x_j|^p \right)^{1/p}.$$

Thus, we can bound $\|\mathbf{Ax}\|_p^p$ as follows,

$$\begin{aligned} \|\mathbf{Ax}\|_p^p &\leq \sum_{i=1}^n \left(\sum_{j=1}^n |A_{ij}| \right)^{p-1} \left(\sum_{j=1}^n |A_{ij}| |x_j|^p \right) \\ &\leq \left(\max_{1 \leq i \leq n} \sum_{j=1}^n |A_{ij}| \right)^{p-1} \sum_{i=1}^n \sum_{j=1}^n |A_{ij}| |x_j|^p \\ &= \|\mathbf{A}\|_\infty^{p-1} \sum_{j=1}^n |x_j|^p \left(\sum_{i=1}^n |A_{ij}| \right) \\ &\leq \|\mathbf{A}\|_\infty^{p-1} \left(\max_{1 \leq j \leq n} \sum_{i=1}^n |A_{ij}| \right) \sum_{j=1}^n |x_j|^p \\ &= \|\mathbf{A}\|_\infty^{p-1} \|\mathbf{A}\|_1 \|x\|_p^p. \end{aligned}$$

The second and fourth inequalities are obtained by taking the max term out of the summation¹, i.e,

$$\sum_{j=1}^n |A_{ij}| \leq \max_{1 \leq i \leq n} \sum_{j=1}^n |A_{ij}| \text{ for all } i, \quad \sum_{i=1}^n |A_{ij}| \leq \max_{1 \leq j \leq n} \sum_{i=1}^n |A_{ij}| \text{ for all } j \quad (8)$$

The third and fifth equalities follow from (a) and (b). Taking the p -th root and maximizing over $\|x\|_p = 1$, we get the inequality

$$\|\mathbf{A}\|_p \leq \|\mathbf{A}\|_\infty^{1-1/p} \|\mathbf{A}\|_1^{1/p}.$$

□

¹This inequality will be used in the proof of Proposition 2 as well.

附录A

附录A.1 命题1的证明

(a) 和 (b) 源于矩阵分析中的标准结果 (Golub & Van Loan, 2013)。

命题1(c)的证明。 令 $\mathbf{A} \in \mathbb{R}^{n \times n}$ 。对于 $p = 1$ 和 $p = \infty$ ，该恒等式显然成立，因此固定 $1 < p < \infty$ 。对于任意 $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{Ax}\|_p^p = \sum_{i=1}^n |(\mathbf{Ax})_i|^p,$$

其中 $(\mathbf{Ax})_i$ 表示 i^{th} 中的 $\mathbf{Ax}$ 元素。然后，对于每个 i ，三角不等式给出

$$|(\mathbf{Ax})_i| = \left| \sum_{j=1}^n A_{ij}x_j \right| \leq \sum_{j=1}^n |A_{ij}| |x_j|$$

根据Hölder不等式，我们得到对于每个 i ,

$$\sum_{j=1}^n |A_{ij}| |x_j| \leq \left(\sum_{j=1}^n |A_{ij}| \right)^{1-1/p} \left(\sum_{j=1}^n |A_{ij}| |x_j|^p \right)^{1/p}.$$

因此，我们可以将 $\|\mathbf{Ax}\|_p^p$ 如下界定，

$$\begin{aligned} \|\mathbf{Ax}\|_p^p &\leq \sum_{i=1}^n \left(\sum_{j=1}^n |A_{ij}| \right)^{p-1} \left(\sum_{j=1}^n |A_{ij}| |x_j|^p \right) \\ &\leq \left(\max_{1 \leq i \leq n} \sum_{j=1}^n |A_{ij}| \right)^{p-1} \sum_{i=1}^n \sum_{j=1}^n |A_{ij}| |x_j|^p \\ &= \|\mathbf{A}\|_\infty^{p-1} \sum_{j=1}^n |x_j|^p \left(\sum_{i=1}^n |A_{ij}| \right) \\ &\leq \|\mathbf{A}\|_\infty^{p-1} \left(\max_{1 \leq j \leq n} \sum_{i=1}^n |A_{ij}| \right) \sum_{j=1}^n |x_j|^p \\ &= \|\mathbf{A}\|_\infty^{p-1} \|\mathbf{A}\|_1 \|x\|_p^p. \end{aligned}$$

第二和第四个不等式是通过从求和¹，中取出最大项得到的，即，

$$\sum_{j=1}^n |A_{ij}| \leq \max_{1 \leq i \leq n} \sum_{j=1}^n |A_{ij}| \text{ for all } i, \quad \sum_{i=1}^n |A_{ij}| \leq \max_{1 \leq j \leq n} \sum_{i=1}^n |A_{ij}| \text{ for all } j \quad (8)$$

第三个和第五个等式源自 (a) 和 (b)。取 p -次方根并在 $\|x\|_p = 1$ 上最大化，我们得到这个不等式

$$\|\mathbf{A}\|_p \leq \|\mathbf{A}\|_\infty^{1-1/p} \|\mathbf{A}\|_1^{1/p}.$$

□

¹这个不等式也将用于命题2的证明中。

A.2 Proof of Theorem 1

Proof of Theorem 1(a). Note that $\mathbf{J}_{\sigma_1}(\mathbf{x})$ is symmetric for all $\mathbf{x} \in \mathbb{R}^n$. Hence, from Proposition 1(a) and (b), we get that for all $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_1 = \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |[\mathbf{J}_{\sigma_1}(\mathbf{x})]_{ij}|$$

From Lemma 2, the above optimization problem becomes,

$$\begin{aligned} & \max_{1 \leq i \leq n} s_i(1 - s_i) + \sum_{j \neq i} s_i s_j \\ &= \max_{1 \leq i \leq n} s_i(1 - s_i) - s_i^2 + \sum_{j=1}^n s_i s_j \\ &= \max_{1 \leq i \leq n} 2s_i(1 - s_i), \end{aligned}$$

where $\mathbf{s} = \sigma_1(\mathbf{x}) = (s_1, s_2, \dots, s_n)$. The optimal value for the above problem is upper bounded by 1/2, and 1/2 is attained for all $\mathbf{x} \in \mathbb{R}^n$ such that the corresponding output $\mathbf{s} = \sigma_1(\mathbf{x})$ belongs to Δ_n^0 and $s_i = 1/2$ for any $i \in \{1, 2, \dots, n\}$. Thus, we get the following bounds,

$$\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_1 \leq \frac{1}{2} \quad \text{and} \quad \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_\infty \leq \frac{1}{2}$$

By Proposition 1(c), we get

$$\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p \leq \left(\frac{1}{2}\right)^{\frac{1}{p}} \left(\frac{1}{2}\right)^{1-\frac{1}{p}} = \frac{1}{2}, \quad \text{for all } 1 \leq p \leq \infty.$$

□

Proof of Theorem 1(b). From Theorem 1(a), we get,

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p \leq \frac{1}{2}, \quad \text{for all } 1 \leq p \leq \infty.$$

From Lemma 3(b), for any $\lambda > 0$, we have,

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_\lambda}(\mathbf{x})\|_p \leq \frac{\lambda}{2} \quad \text{for all } 1 < p < \infty.$$

and hence using Lipschitz characterization in Lemma 1,

$$\|\sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y})\|_p \leq \frac{\lambda}{2} \|\mathbf{x} - \mathbf{y}\|_p \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n \text{ and for all } 1 \leq p \leq \infty.$$

□

A.3 Proof of Proposition 2

Define $\mathbf{M}(\mathbf{s}) \in \mathbb{R}^{n \times n}$ as $\text{diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^T$. From Lemma 3, for all $1 \leq p \leq \infty$,

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p = \sup_{\mathbf{s} \in \Delta_n^\circ} \|\mathbf{M}(\mathbf{s})\|_p$$

Proof of Proposition 2(a). As discussed in Theorem 1(a), for $p = 1$ and $p = \infty$, for all $\mathbf{s} \in \Delta_n^\circ$,

$$\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = \max_{1 \leq i \leq n} 2s_i(1 - s_i) \leq 1/2$$

A.2 定理1的证明

定理**1(a)**的证明。注意 $\mathbf{J}_{\sigma_1}(\mathbf{x})$ 对所有 $\mathbf{x} \in \mathbb{R}^n$ 都是对称的。因此，根据命题1(a)和(b)，我们得到对所有 $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_1 = \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |[\mathbf{J}_{\sigma_1}(\mathbf{x})]_{ij}|$$

由引理2，上述优化问题变为，

$$\begin{aligned} & \max_{1 \leq i \leq n} s_i(1 - s_i) + \sum_{j \neq i} s_i s_j \\ &= \max_{1 \leq i \leq n} s_i(1 - s_i) - s_i^2 + \sum_{j=1}^n s_i s_j \\ &= \max_{1 \leq i \leq n} 2s_i(1 - s_i), \end{aligned}$$

其中 $\mathbf{s} = \sigma_1(\mathbf{x}) = (s_1, s_2, \dots, s_n)$. 该问题的最优值被 1/2 上界，且1/2 对于所有满足对应输出 $\mathbf{s} = \sigma_1(\mathbf{x})$ 属于 Δ_n^0 和 $s_i = 1/2$ 的 $\mathbf{x} \in \mathbb{R}^n$ 都成立，其中 $i \in \{1, 2, \dots, n\}$. 因此，我们得到以下界限，

$$\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_1 \leq \frac{1}{2} \quad \text{and} \quad \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_\infty \leq \frac{1}{2}$$

由命题1(c)，我们得到

$$\|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p \leq \left(\frac{1}{2}\right)^{\frac{1}{p}} \left(\frac{1}{2}\right)^{1-\frac{1}{p}} = \frac{1}{2}, \quad \text{for all } 1 \leq p \leq \infty.$$

□

定理**1(b)**的证明。由定理1(a)，我们得到，

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p \leq \frac{1}{2}, \quad \text{for all } 1 \leq p \leq \infty.$$

从引理3(b)可知，对于任意 $\lambda > 0$ ，我们有，

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_\lambda}(\mathbf{x})\|_p \leq \frac{\lambda}{2} \quad \text{for all } 1 < p < \infty.$$

因此，利用引理1中的Lipschitz特性，

$$\|\sigma_\lambda(\mathbf{x}) - \sigma_\lambda(\mathbf{y})\|_p \leq \frac{\lambda}{2} \|\mathbf{x} - \mathbf{y}\|_p \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n \text{ and for all } 1 \leq p \leq \infty.$$

□

A.3 命题2的证明

定义 $\mathbf{M}(\mathbf{s}) \in \mathbb{R}^{n \times n}$ 为 $\text{diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^T$ 。根据引理3，对于所有 $1 \leq p \leq \infty$,

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_p = \sup_{\mathbf{s} \in \Delta_n^\circ} \|\mathbf{M}(\mathbf{s})\|_p$$

命题**2(a)**的证明。如定理1(a)所述，对于 $p = 1$ 和 $p = \infty$ ，对于所有 $\mathbf{s} \in \Delta_n^\circ$,

$$\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = \max_{1 \leq i \leq n} 2s_i(1 - s_i) \leq 1/2$$

and $1/2$ is attained if there exists $s \in \Delta_n^\circ$ such that $s_i = 1/2$ for some i and $s_j > 0$ for $j \neq i$.

To prove that $1/2$ is indeed attained, we need to show an example where $1/2$ is attained at a point $\mathbf{s}$ in the interior of the probability simplex Δ_n° .

Let $\mathbf{x} \in \mathbb{R}^n$ such that $x_i = \ln(n-1)$ and $x_j = 0$ for all $j \neq i$, $\mathbf{s} = \sigma_1(\mathbf{x})$ is such that $s_i = 1/2$ and $s_j > 0$ for all $j \neq i$ and hence $\mathbf{s} \in \Delta_n^\circ$. Thus,

$$\max_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_1 = \max_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_\infty = 1/2$$

□

Proof of Proposition 2(b). We first prove for the case $n > 2$. It suffices to show that for arbitrary $1 < p < \infty$, $\|\mathbf{M}(\mathbf{s})\|_p \neq 1/2$ for any $\mathbf{s} \in \Delta_n^\circ$ and there exists $\mathbf{s} \in \partial\Delta_n$ such that $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ for $n > 2$ to prove that $1/2$ is indeed approached in limit. We need the following definition of the support of a vector and the corresponding lemma to prove this.

Definition A.1 (Support of a vector). For a vector $\mathbf{s} \in \mathbb{R}^n$, the support of $\mathbf{s}$ is defined as

$$\text{supp}(\mathbf{s}) := \{i \in \{1, \dots, n\} : s_i \neq 0\}.$$

Lemma 4 (Support condition for attaining $1/2$). Let $\mathbf{M}(\mathbf{s}) \in \mathbb{R}^{n \times n}$ be defined by

$$\mathbf{M}(\mathbf{s}) := \text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top,$$

for $\mathbf{s} \in \Delta_n$. Suppose $1 < p < \infty$ and $n > 2$. Then, $\|\mathbf{M}(\mathbf{s})\|_p = \frac{1}{2}$ if and only if $\mathbf{s}$ is a permutation of $(1/2, 1/2, 0, \dots, 0)$

The proof of the lemma is given towards the end of this section. From Lemma 4, there exists no $\mathbf{s} \in \Delta_n$ with $\text{supp}(\mathbf{s}) > 2$ with $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$. Hence, there exists no $\mathbf{s} \in \Delta_n^\circ$ with $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ since $\text{supp}(\mathbf{s}) = n$ for all $\mathbf{s} \in \Delta_n^\circ$. Also, there exists $\mathbf{s}$ with $\text{supp}(\mathbf{s}) = 2$ such that $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$.

Let $\hat{\mathbf{s}} \in \Delta_n$ with $\text{supp}(\hat{\mathbf{s}}) = 2$ be such that $\|\mathbf{M}(\hat{\mathbf{s}})\|_p = 1/2$. By definition, $\hat{\mathbf{s}} \in \partial\Delta_n$ (boundary of the simplex). Hence, we can obtain a sequence of $\mathbf{s}_k \in \Delta_n^\circ$ (interior of the simplex) such that $\mathbf{s}_k \rightarrow \hat{\mathbf{s}}$ as $k \rightarrow \infty$ (Rockafellar & Wets, 1998). Also since $\mathbf{M}(\mathbf{s})$ is a continuous function, $\mathbf{M}(\mathbf{s}_k) \rightarrow 1/2$ as $k \rightarrow \infty$. For completeness, we show convergence by constructing example sequence in Example 2.

Finally, when $n = 2$, the point $\mathbf{s} = (1/2, 1/2)$ lies in Δ_2° and attains $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$. □

Example 2 (Sequence $(\mathbf{s}_k)_{k \geq 1} \subset \Delta_n^\circ$ such that $\mathbf{M}(\mathbf{s}_k) \rightarrow 1/2$ as $k \rightarrow \infty$). Consider $\hat{\mathbf{s}} \in \partial\Delta_n$ with $\hat{s}_1 = \hat{s}_2 = 1/2$ and all other entries zero. By Lemma 4, $\|\mathbf{M}(\hat{\mathbf{s}})\|_p = 1/2$ To approximate $\hat{\mathbf{s}}$ from the interior of the probability simplex, fix $\epsilon \in (0, 1/2)$ and choose $\delta \in (0, 1/2)$ satisfying $2\delta(1 - \delta) = \epsilon$. Define

$$s_1 = s_2 = \frac{1}{2} - \delta, \quad s_j = \frac{2\delta}{n-2}, \quad j \geq 3,$$

so that $\mathbf{s} \in \Delta_n^\circ$. Consider the vector

$$\mathbf{v} = (2^{-1/p}, -2^{-1/p}, 0, \dots, 0) \in \mathbb{R}^n \quad \text{with} \quad \|\mathbf{v}\|_p = 1,$$

A direct computation shows,

$$[\mathbf{M}(\mathbf{s})\mathbf{v}]_1 = -[\mathbf{M}(\mathbf{s})\mathbf{v}]_2 = \frac{2}{2^{1/p}} \left(\frac{1}{2} - \delta \right)^2 \quad \text{and} \quad [\mathbf{M}(\mathbf{s})\mathbf{v}]_j = 0 \quad \text{for all } j \notin \{1, 2\}.$$

Hence, we get,

$$\|\mathbf{M}(\mathbf{s})\mathbf{v}\|_p = \frac{1}{2} - 2\delta(1 - \delta) = \frac{1}{2} - \epsilon.$$

Thus for every $\epsilon \in (0, 1/2)$, there exists $\mathbf{s} \in \Delta_n^\circ$ with $1/2 - \epsilon \leq \|\mathbf{M}(\mathbf{s})\|_p < 1/2$, proving that $1/2$ is the tight upper bound.

并且 $1/2$ 被达到, 如果存在 $s \in \Delta_n^\circ$, 使得 $s_i = 1/2$ 对于某些 i 并且 $s_j > 0$ 对于 $j \neq i$.

为证明 $1/2$ 确实被达到, 我们需要展示一个例子, 其中 $1/2$ 在概率单纯形 Δ_n° 的内部点 $\mathbf{s}$ 处被达到。

设 $\mathbf{x} \in \mathbb{R}^n$ 满足 $x_i = \ln(n-1)$ 和 $x_j = 0$ 对所有 $j \neq i$, $\mathbf{s} = \sigma_1(\mathbf{x})$ 成立, 使得 $s_i = 1/2$ 和 $s_j > 0$ 对所有 $j \neq i$ 成立, 从而 $\mathbf{s} \in \Delta_n^\circ$. 因此,

$$\max_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_1 = \max_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{J}_{\sigma_1}(\mathbf{x})\|_\infty = 1/2$$

□

命题**2(b)**的证明。我们首先对 $n > 2$ 的情况进行证明。只需证明对于任意的 $1 < p < \infty$, $\|\mathbf{M}(\mathbf{s})\|_p \neq 1/2$ 对任何 $\mathbf{s} \in \Delta_n^\circ$ 成立, 并且存在 $\mathbf{s} \in \partial\Delta_n$ 使得 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ 对 $n > 2$ 成立, 以证明 $1/2$ 确实在极限中被逼近。我们需要以下向量的支撑定义和相应的引理来证明这一点。

定义 A.1 (向量的支撑集). 对于向量 $\mathbf{s} \in \mathbb{R}^n$, $\mathbf{s}$ 的支撑集定义为 as

$$\text{supp}(\mathbf{s}) := \{i \in \{1, \dots, n\} : s_i \neq 0\}.$$

引理4 (达到 $1/2$ 的支撑条件). 令 $\mathbf{M}(\mathbf{s}) \in \mathbb{R}^{n \times n}$ 被定义为

$$\mathbf{M}(\mathbf{s}) := \text{Diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top,$$

对于 $\mathbf{s} \in \Delta_n$. 假设 $1 < p < \infty$ 和 $n > 2$. 那么, $\|\mathbf{M}(\mathbf{s})\|_p = \frac{1}{2}$ 当且仅当 $\mathbf{s}$ 是 $(1/2, 1/2, 0, \dots, 0)$ 的排列

该引理的证明在本节末尾给出。根据引理4, 不存在 $\mathbf{s} \in \Delta_n$ 满足 $\text{supp}(\mathbf{s}) > 2$ 满足 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$. 因此, 不存在 $\mathbf{s} \in \Delta_n^\circ$ 满足 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$, 因为对于所有 $\mathbf{s} \in \Delta_n^\circ$, $\text{supp}(\mathbf{s}) = n$. 此外, 存在 $\mathbf{s}$ 满足 $\text{supp}(\mathbf{s}) = 2$, 使得 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$.

设 $\hat{\mathbf{s}} \in \Delta_n$ 的支撑集 $\hat{\mathbf{s}}$ 包含 $= 2$, 且满足 $\|\mathbf{M}(\hat{\mathbf{s}})\|_p = 1/2$. 根据定义, $\hat{\mathbf{s}} \in \partial\Delta_n$ 是概率单纯形的边界。因此, 我们可以得到一个序列 $\mathbf{s}_k \in \Delta_n^\circ$ (单纯形内部) 使得 $\mathbf{s}_k$ 收敛到 $k \rightarrow \infty$ (Rockafellar & Wets, 1998)。此外, 由于 $\mathbf{M}(\mathbf{s})$ 是连续函数, $\mathbf{M}(\mathbf{s}_k)$ 收敛到 $\rightarrow 1/2$ 。为了完整性, 我们在示例2中通过构造一个例子来证明收敛性。

最后, 当 $n = 2$ 时, 点 $\mathbf{s} = (1/2, 1/2)$ 位于 Δ_2° 并达到 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$. □

示例2 (序列 $(\mathbf{s}_k)_{k \geq 1} \subset \Delta_n^\circ$ 使得 $\mathbf{M}(\mathbf{s}_k)$ 收敛到 $k \rightarrow \infty$)。考虑 $\hat{\mathbf{s}} \in \partial\Delta_n$, 其中 $\hat{s}_1 = \hat{s}_2 = 1/2$, 其余项为零。根据引理4, $\|\mathbf{M}(\hat{\mathbf{s}})\|_p = 1/2$ 。为了从概率单纯形内部逼近 $\hat{\mathbf{s}}$, 固定 $\epsilon \in (0, 1/2)$ 并选择 $\delta \in (0, 1/2)$ 满足 $2\delta(1 - \delta) = \epsilon$ 。定义

$$s_1 = s_2 = \frac{1}{2} - \delta, \quad s_j = \frac{2\delta}{n-2}, \quad j \geq 3,$$

以便 $\mathbf{s} \in \Delta_n^\circ$ 。考虑向量

$$\mathbf{v} = (2^{-1/p}, -2^{-1/p}, 0, \dots, 0) \in \mathbb{R}^n \quad \text{with} \quad \|\mathbf{v}\|_p = 1,$$

直接计算表明,

$$[\mathbf{M}(\mathbf{s})\mathbf{v}]_1 = -[\mathbf{M}(\mathbf{s})\mathbf{v}]_2 = \frac{2}{2^{1/p}} \left(\frac{1}{2} - \delta \right)^2 \quad \text{and} \quad [\mathbf{M}(\mathbf{s})\mathbf{v}]_j = 0 \quad \text{for all } j \notin \{1, 2\}.$$

因此, 我们得到,

$$\|\mathbf{M}(\mathbf{s})\mathbf{v}\|_p = \frac{1}{2} - 2\delta(1 - \delta) = \frac{1}{2} - \epsilon.$$

因此对于每一个 $\epsilon \in (0, 1/2)$, 都存在 $\mathbf{s} \in \Delta_n^\circ$ 满足 $1/2 - \epsilon \leq \|\mathbf{M}(\mathbf{s})\|_p < 1/2$, 证明 $1/2$ 是紧的上界。

Proof of Lemma 4. Let $\mathbf{s}$ be a permutation of $(1/2, 1/2, 0, \dots, 0)$ i.e $s_i = s_j = 1/2$ for some distinct $i, j \in \{1, 2, \dots, n\}$. $\mathbf{M}(\mathbf{s})$ has values $\mathbf{M}(\mathbf{s})_{ii} = \mathbf{M}(\mathbf{s})_{jj} = 1/4$, $\mathbf{M}(\mathbf{s})_{ij} = \mathbf{M}(\mathbf{s})_{ji} = -1/4$, and zeros elsewhere. To show that $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$, it suffices to construct $\mathbf{v} \in \mathbb{R}^n$ such that $\|\mathbf{v}\|_p = 1$ and $\|\mathbf{M}(\mathbf{s})\mathbf{v}\|_p = 1/2$. Consider $\mathbf{v}$ to be a similar permutation of

$$(2^{-1/p}, -2^{-1/p}, 0, \dots, 0) \in \mathbb{R}^n.$$

i.e. $v_i = 2^{-1/p}$ and $v_j = -2^{-1/p}$. A direct calculation gives $\|\mathbf{M}(\mathbf{s})\mathbf{v}\|_p = 1/2$.

Now we need to prove that if $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$, $\mathbf{s}$ is a permutation of $(1/2, 1/2, 0, \dots, 0)$. By Proposition 1(c), the interpolation inequality gives

$$\|\mathbf{M}(\mathbf{s})\|_p \leq \|\mathbf{M}(\mathbf{s})\|_1^{1/p} \|\mathbf{M}(\mathbf{s})\|_\infty^{1-1/p}.$$

From Proposition 2(a), we know $\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty \leq 1/2$. Thus if $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$, equality must hold throughout. This means for all $\mathbf{s} \in \Delta_n$ such that $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$, we have $\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = 1/2$.

We divide into cases according to the support size of $\mathbf{s}$.

Case (i): $\text{supp}(\mathbf{s}) = 1$. Suppose $s_i = 1$ for some i , and $s_j = 0$ for all $j \neq i$. Then $\mathbf{M}(\mathbf{s}) = 0$. Hence $\|\mathbf{M}(\mathbf{s})\|_p = 0$ for all p . Thus, the value $1/2$ cannot be attained for any $\mathbf{s}$.

Case (ii): $\text{supp}(\mathbf{s}) = 2$. As discussed in Theorem 1(a), for $p = 1$ and $p = \infty$, for all $\mathbf{s} \in \Delta_n$,

$$\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = \max_{1 \leq i \leq n} 2s_i(1 - s_i) \leq 1/2$$

and $1/2$ is attained only if there exists $\mathbf{s} \in \Delta_n$ such that $s_i = 1/2$ for some i . Thus, if $\text{supp}(\mathbf{s}) = 2$, then $\mathbf{s}$ needs to be permutations of $(1/2, 1/2, 0, \dots, 0)$ for $\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = 1/2$ and we have already shown $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ for any permutation of $(1/2, 1/2, 0, \dots, 0)$.

Thus, if $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ with $\text{supp}(\mathbf{s}) = 2$, then $\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = 1/2$ which needs $\mathbf{s}$ to be a permutation of $(1/2, 1/2, 0, \dots, 0)$.

Case (iii): $\text{supp}(\mathbf{s}) > 2$. Suppose, $\hat{\mathbf{s}} \in \Delta_n$ with $\text{supp}(\hat{\mathbf{s}}) > 2$ satisfies $\|\mathbf{M}(\hat{\mathbf{s}})\|_p = 1/2$. By Proposition 1(c),

$$\|\mathbf{M}(\hat{\mathbf{s}})\|_p \leq \|\mathbf{M}(\hat{\mathbf{s}})\|_1^{1/p} \|\mathbf{M}(\hat{\mathbf{s}})\|_\infty^{1-1/p}.$$

Since $\|\mathbf{M}(\hat{\mathbf{s}})\|_1 = \|\mathbf{M}(\hat{\mathbf{s}})\|_\infty = 1/2$ holds, equality must hold in the interpolation inequality in Proposition 1. Thus, it suffices to rule out equality.

We just need to show that one of the inequalities used in the proof of Proposition 1(c) is strict. We show that the inequality in Eq. 8 is strict i.e. there exists $1 \leq i \leq n$, such that,

$$\sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{ij}| < \max_{1 \leq k \leq n} \sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{kj}|$$

Since $\|\mathbf{M}(\hat{\mathbf{s}})\|_1 = \|\mathbf{M}(\hat{\mathbf{s}})\|_\infty = 1/2$, $\hat{s}_{i_1} = 1/2$ for some $1 \leq i_1 \leq n$ and $\hat{s}_j < 1/2$ for all $j \neq i_1$. Since $\text{supp}(\hat{\mathbf{s}}) > 2$, there exists an index $i_2 \in \{1, 2, \dots, n\}$ and $i_2 \neq i_1$, where $\hat{s}_{i_2} < 1/2$. As discussed in Theorem 1, for any row index i , $\sum_{j=1}^n |\mathbf{M}(\hat{\mathbf{s}})_{ij}| = 2\hat{s}_i(1 - \hat{s}_i)$. Hence,

$$\max_{1 \leq i \leq n} \sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{ij}| = \sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{i_1 j}| = 1/2,$$

and for row index i_2 ,

$$\sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{i_2 j}| < 1/2 = \max_{1 \leq k \leq n} \sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{kj}|.$$

This shows that the inequality in Eq. 8 is strict. Hence, the interpolation inequality in Proposition 1(c) is strict for the matrix $\mathbf{M}(\hat{\mathbf{s}})$. This contradicts our assumption that $\|\mathbf{M}(\hat{\mathbf{s}})\|_p = 1/2$.

Combining all cases, we conclude that $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ if and only if $\mathbf{s}$ is a permutation of $(1/2, 1/2, 0, \dots, 0)$. $\square$

引理4的证明. 设 $\mathbf{s}$ 是 $(1/2, 1/2, 0, \dots, 0)$ 的一个排列, 即 $s_i = s_j = 1/2$ 对于某些不同的 $i, j \in \{1, 2, \dots, n\}$ 。

$\mathbf{M}(\mathbf{s})$ 的值为 $\mathbf{M}(\mathbf{s})_{ii} = \mathbf{M}(\mathbf{s})_{jj} = 1/4$, $\mathbf{M}(\mathbf{s})_{ij} = \mathbf{M}(\mathbf{s})_{ji} = -1/4$, 其余为零。为证明 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$, 只需构造 $\mathbf{v} \in \mathbb{R}^n$, 使得 $\|\mathbf{v}\|_p = 1$ 且 $\|\mathbf{M}(\mathbf{s})\mathbf{v}\|_p = 1/2$ 。考虑将 $\mathbf{v}$ 视为一个类似的排列

$$(2^{-1/p}, -2^{-1/p}, 0, \dots, 0) \in \mathbb{R}^n.$$

即 $v_i = 2^{-1/p}$ 和 $v_j = -2^{-1/p}$ 。直接计算得到 $\|\mathbf{M}(\mathbf{s})\mathbf{v}\|_p = 1/2$ 。

现在我们需要证明: 如果 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$, $\mathbf{s}$ 是 $(1/2, 1/2, 0, \dots, 0)$ 的一个排列。根据命题1(c), 插值不等式给出

$$\|\mathbf{M}(\mathbf{s})\|_p \leq \|\mathbf{M}(\mathbf{s})\|_1^{1/p} \|\mathbf{M}(\mathbf{s})\|_\infty^{1-1/p}.$$

根据命题2(a), 我们知道 $\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty \leq 1/2$ 。因此, 如果 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$, 则整个式子必须相等。这意味着对于所有满足 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ 的 $\mathbf{s} \in \Delta_n$, 我们有 $\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = 1/2$ 。

我们根据 $\mathbf{s}$ 的支撑大小分为情况。

情况(i): $\text{supp}(\mathbf{s}) = 1$ 。假设 $s_i = 1$ 对某些 i 成立, 并且 $s_j = 0$ 对所有 $j \neq i$ 成立。那么 $\mathbf{M}(\mathbf{s}) = 0$ 。因此 $\|\mathbf{M}(\mathbf{s})\|_p = 0$ 对所有 p 成立。因此, 值 $1/2$ 对任何 $\mathbf{s}$ 都无法达到。

情况(ii): $\text{supp}(\mathbf{s}) = 2$ 。正如定理1(a)中讨论的, 对于 $p = 1$ 和 $p = \infty$, 对于所有 $\mathbf{s} \in \Delta_n$,

$$\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = \max_{1 \leq i \leq n} 2s_i(1 - s_i) \leq 1/2$$

并且 $1/2$ 只有在存在 $\mathbf{s} \in \Delta_n$ 使得 $s_i = 1/2$ 对某些 i 成立时才会达到。因此, 如果 $\text{supp}(\mathbf{s}) = 2$, 那么 $\mathbf{s}$ 需要是 $(1/2, 1/2, 0, \dots, 0)$ 的排列, 对于 $\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = 1/2$, 并且我们已经证明了 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ 对 $(1/2, 1/2, 0, \dots, 0)$ 的任何排列都成立。

因此, 如果 $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ 在 $\mathbf{s}$ 的支撑集上 $= 2$, 那么 $\|\mathbf{M}(\mathbf{s})\|_1 = \|\mathbf{M}(\mathbf{s})\|_\infty = 1/2$, 这需要 $\mathbf{s}$ 是 $(1/2, 1/2, 0, \dots, 0)$ 的排列。

情况 (iii): $\text{supp}(\mathbf{s}) > 2$ 。假设, $\hat{\mathbf{s}} \in \Delta_n$ 在 $\mathbf{s}$ 的支撑集上 > 2 满足 $\|\mathbf{M}(\hat{\mathbf{s}})\|_p = 1/2$ 。根据命题1(c),

$$\|\mathbf{M}(\hat{\mathbf{s}})\|_p \leq \|\mathbf{M}(\hat{\mathbf{s}})\|_1^{1/p} \|\mathbf{M}(\hat{\mathbf{s}})\|_\infty^{1-1/p}.$$

由于 $\|\mathbf{M}(\hat{\mathbf{s}})\|_1 = \|\mathbf{M}(\hat{\mathbf{s}})\|_\infty = 1/2$ 成立, 命题1中的插值不等式必须取等号。因此, 只需排除等号成立的情况。

我们只需证明命题1(c)的证明中使用的其中一个不等式是严格的。我们证明公式8中的不等式是严格的, 即存在 $1 \leq i \leq n$, 使得,

$$\sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{ij}| < \max_{1 \leq k \leq n} \sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{kj}|$$

由于 $\|\mathbf{M}(\hat{\mathbf{s}})\|_1 = \|\mathbf{M}(\hat{\mathbf{s}})\|_\infty = 1/2$, $\hat{s}_{i_1} = 1/2$ 对某些 $1 \leq i_1 \leq n$ 成立, 且 $\hat{s}_j < 1/2$ 对所有 $j \neq i_1$ 成立。由于 $\text{supp}(\hat{\mathbf{s}}) > 2$, 存在一个索引 $i_2 \in \{1, 2, \dots, n\}$ 和 $i_2 \neq i_1$, 其中 $\hat{s}_{i_2} < 1/2$ 。正如定理1中所讨论的, 对于任何行索引 i , $\sum_{j=1}^n |\mathbf{M}(\hat{\mathbf{s}})_{ij}| = 2\hat{s}_i(1 - \hat{s}_i)$ 。因此,

$$\max_{1 \leq i \leq n} \sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{ij}| = \sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{i_1 j}| = 1/2,$$

对于行索引 i_2 ,

$$\sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{i_2 j}| < 1/2 = \max_{1 \leq k \leq n} \sum_{j=1}^n |\mathbf{M}(\mathbf{s})_{kj}|.$$

这表明公式8中的不等式是严格的。因此, 对于矩阵 $\mathbf{M}(\hat{\mathbf{s}})$, 命题1(c)中的插值不等式是严格的。这与我们假设的 $\|\mathbf{M}(\hat{\mathbf{s}})\|_p = 1/2$ 相矛盾。

综合所有情况, 我们得出结论: $\|\mathbf{M}(\mathbf{s})\|_p = 1/2$ 当且仅当 $\mathbf{s}$ 是 $(1/2, 1/2, 0, \dots, 0)$ 的一个排列。 $\square$

A.4 Proof of Theorem 3

Let the DSFP map be defined as

$$T(\mathbf{y}) = \sigma_{1/\tau}(\mathbf{A}^\top \sigma_{1/\tau}(-\mathbf{A}\mathbf{y})), \quad \mathbf{y} \in \Delta_m,$$

where $\sigma_{1/\tau}$ denotes the softmax operator with temperature τ . By Theorem 1, the softmax operator is globally Lipschitz with constant $\frac{1}{2\tau}$, uniformly across all ℓ_p norms with $1 \leq p \leq \infty$. Applying the chain rule of Lipschitz constants, we obtain

$$\|T(\mathbf{y}_1) - T(\mathbf{y}_2)\|_p \leq \frac{1}{2\tau} \|\mathbf{A}^\top\|_p \frac{1}{2\tau} \|\mathbf{A}\|_p \|\mathbf{y}_1 - \mathbf{y}_2\|_p.$$

Since $\|\mathbf{A}^\top\|_p = \|\mathbf{A}\|_p$, this yields

$$\|T(\mathbf{y}_1) - T(\mathbf{y}_2)\|_p \leq \frac{\|\mathbf{A}\|_p^2}{4\tau^2} \|\mathbf{y}_1 - \mathbf{y}_2\|_p.$$

Hence, the DSFP map has the Lipschitz constant,

$$c = \frac{\|\mathbf{A}\|_p^2}{4\tau^2}.$$

If $\tau > \|\mathbf{A}\|_p/2$, then $c < 1$ and T is a Banach contraction on the complete metric space $(\Delta_m, \|\cdot\|_p)$. By the Banach fixed-point theorem, T then admits a unique fixed point $\mathbf{y}^* \in \Delta_m$, and for any initialization $\mathbf{y}_0 \in \Delta_m$, the iteration

$$\mathbf{y}_{k+1} \leftarrow (1 - \alpha) \mathbf{y}_k + \alpha T(\mathbf{y}_k), \quad \alpha \in (0, 1],$$

converges linearly in the ℓ_p norm to $\mathbf{y}^*$. At each step, the row player's strategy is updated as

$$\mathbf{x}_k = \sigma_{1/\tau}(-\mathbf{A}\mathbf{y}_k).$$

Since $\mathbf{y}_k \rightarrow \mathbf{y}^*$ and the softmax map is continuous, it follows that $\mathbf{x}_k \rightarrow \mathbf{x}^* := \sigma_{1/\tau}(-\mathbf{A}\mathbf{y}^*)$. Thus, both players' equilibrium strategies are recovered in the limit.

A.4 定理3的证明

设DSFP映射定义为

$$T(\mathbf{y}) = \sigma_{1/\tau}(\mathbf{A}^\top \sigma_{1/\tau}(-\mathbf{A}\mathbf{y})), \quad \mathbf{y} \in \Delta_m,$$

其中 $\sigma_{1/\tau}$ 表示温度为 τ 的softmax算子。根据定理1, softmax算子在整个 $\mathbb{R}^n$ 空间中是利普希茨连续的, 利普希茨常数为 $\frac{1}{2\tau}$, 并且对于所有满足 $1 \leq p \leq \infty$ 的 ℓ_p 范数都成立。应用利普希茨常数的链式法则, 我们得到

$$\|T(\mathbf{y}_1) - T(\mathbf{y}_2)\|_p \leq \frac{1}{2\tau} \|\mathbf{A}^\top\|_p \frac{1}{2\tau} \|\mathbf{A}\|_p \|\mathbf{y}_1 - \mathbf{y}_2\|_p.$$

由于 $\|\mathbf{A}^\top\|_p = \|\mathbf{A}\|_p$, 因此得到

$$\|T(\mathbf{y}_1) - T(\mathbf{y}_2)\|_p \leq \frac{\|\mathbf{A}\|_p^2}{4\tau^2} \|\mathbf{y}_1 - \mathbf{y}_2\|_p.$$

因此, DSFP映射具有利普希茨常数

$$c = \frac{\|\mathbf{A}\|_p^2}{4\tau^2}.$$

如果 $\tau > \|\mathbf{A}\|_p/2$, 那么 $c < 1$ 并且 T 在完备度量空间 $(\Delta_m, \|\cdot\|_p)$ 上是Banach收缩。根据Banach不动点定理, T 存在唯一的不动点 $\mathbf{y}^* \in \Delta_m$, 并且对于任何初始化 $\mathbf{y}_0 \in \Delta_m$, 迭代

$$\mathbf{y}_{k+1} \leftarrow (1 - \alpha) \mathbf{y}_k + \alpha T(\mathbf{y}_k), \quad \alpha \in (0, 1],$$

收敛在 ℓ_p 范数下线性收敛到 $\mathbf{y}^*$ 。每一步, 行玩家的策略都会更新

$$\mathbf{x}_k = \sigma_{1/\tau}(-\mathbf{A}\mathbf{y}_k).$$

由于 $\mathbf{y}_k \rightarrow \mathbf{y}^*$ 且softmax映射是连续的, 因此可以推出 $\mathbf{x}_k \rightarrow \mathbf{x}^* := \sigma_{1/\tau}(-\mathbf{A}\mathbf{y}^*)$ 。因此, 两位玩家的均衡策略在极限下得以恢复。